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From Research Spaces to Strategic Portfolio Design: Forecasting Country-Level Scholarly Diversification

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Abstract

This paper examines whether research-space analysis can support portfolio design informed by forecasts at the country–subfield level. Using a bibliographic catalog of scientific papers, authors and institutions, we construct a research space from country activity across subfields and test whether measures derived from that space help predict later changes in specialization. The study contributes an empirical framework that links three elements: historical validation of research-space measures, comparison of alternative specifications of relative relatedness, and portfolio selection under an explicit complexity target. We estimate reduced-form stepping-stone models for entry and exit and evaluate them using in-sample evidence, leave-one-year-out checks, and robustness comparisons across relatedness constructions. The results show that research-space measures help predict later specialization changes and that relative relatedness standardized over the opportunity set provides the most consistent specification, especially for entry into previously non-specialized subfields. Using this specification, we formulate a prospective portfolio problem in which candidate subfields are selected to increase portfolio complexity while minimizing predicted entry effort. In a 2024 comparison covering 164 countries, optimized portfolios differ markedly from relatedness rankings, with 92.7% of countries shifting toward lower auxiliary volume and higher predicted future sophistication.

Keywords: economic complexity; specialization dynamics; research space; relative relatedness; portfolio optimization; decision support

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1. Introduction

How countries expand their research portfolios is not only a descriptive question but also a strategic one. New specialization does not emerge at random: it tends to appear in areas that are closer to capabilities already accumulated. This is why relatedness-based approaches to diversification [1,2] have become central to the study of structural change. In the economic-complexity literature, revealed comparative advantage, portfolio

composition and proximity structures help explain why some activities are more feasible than others and why diversification is shaped by inherited capability endowments rather than unconstrained choice [3–8]. Recent extensions have further shown that this logic is not confined to trade and manufacturing, but can also be applied to scientific and technological domains [9].

Within the scholarly domain, this perspective has motivated research-space representations that organize fields, topics, or research areas according to patterns of co-occurrence, career trajectories, or other indicators of knowledge proximity [10,11]. These studies show that science can be represented as a structured opportunity space rather than as a collection of isolated fields. At the same time, bibliometric infrastructures now make it increasingly feasible to analyze such patterns at large scale and across countries in a consistent and reproducible way [12,13]. This use of large-scale data to examine the organization and evolution of scientific activity is also aligned with the broader science-of-science literature [14].

However, a methodological gap remains. Existing research-space studies have constructed scientific opportunity spaces from bibliographic and relational signals, including co-occurrence patterns, career trajectories, shared authorship and field development [10,11]. These approaches make the topology of scientific activity observable and can support descriptive mapping, field interpretation and broad anticipation of future development. Yet, in many applications, the measures derived from these spaces remain separate from a direct historical assessment of later specialization dynamics. In parallel, recent work in economic complexity has shown that, once diversification is formulated as target attainment under feasibility conditions, the activities favored by a relatedness-based ordering need not coincide with those selected under formal optimization criteria [15,16]. In the scholarly domain, these two strands have rarely been integrated within a common country–subfield setting in which research-space measures are first assessed historically and then used to inform target-based portfolio selection.

A related methodological issue concerns the operationalization of relatedness for comparative inference. The relatedness literature has shown that choices regarding normalization, support sets and benchmark definitions can materially affect interpretation and downstream policy use [17]. This issue is especially relevant in scholarly settings, where cross-country comparisons are sensitive to differences in portfolio size, specialization breadth and the composition of the contemporaneous opportunity set. In this context, the way relative relatedness is constructed affects the empirical specification itself.

These strands motivate the integration developed in this article. Studies on relatedness and economic complexity provide the basis for modeling diversification as expansion constrained by accumulated capabilities; research-space studies support the representation of scientific activity as a structured opportunity space; and optimization work clarifies why portfolio selection under an explicit target may differ from heuristic ranking. Consequently, the present study links the historical assessment of measures derived from the research space with prospective portfolio selection in the country–subfield setting.

Specifically, we present an integrated empirical framework that supports decisions about which subfields countries should prioritize when the objective is to raise portfolio sophistication under feasibility constraints. Using publication data, we construct a research space under a common protocol, test whether its signals contain stable predictive information about subsequent entry and exit dynamics, and then use the historically validated benchmark in a prospective portfolio setting with an explicit target criterion. For decision-oriented use, the framework provides a structured way to compare candidate subfields under explicit assumptions about feasibility, predicted sophistication and target attainment. Its outputs are intended to support science policy and research management discussions, rather than to replace expert judgment or institutional priorities.

The article makes the following contributions:

- A country–subfield framework in which measures derived from the research space are evaluated against later specialization changes, rather than used only for descriptive mapping.
- A comparison of alternative specifications of relative relatedness, showing how this modeling choice affects inference and motivating the benchmark specification used in the portfolio analysis.
- A portfolio problem for scholarly diversification in which candidate subfields are selected under an explicit complexity target, rather than through sequential relatedness ranking.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on economic complexity, relatedness, research-space analysis and diversification forecasting. Section 3 presents the empirical design, including data construction, research-space estimation, historical assessment and the prospective portfolio-design layer. Section 4 reports the historical validation results and the prospective portfolio comparisons. Section 5 discusses the methodological and substantive implications of the findings. Section 6 summarizes the conclusions, limitations and future work.

2. Related Work

This section situates the article conceptually and methodologically by connecting three bodies of prior work with the main components of the empirical framework. Section 2.1 reviews economic complexity and relatedness concepts that motivate the measures of specialization, proximity and portfolio position. Section 2.2 examines research-space and scholarly diversification studies that support the representation of scientific activity as a structured opportunity space. Section 2.3 discusses prioritization and optimization approaches that motivate the portfolio layer under an explicit target.

2.1. Economic Complexity, Relatedness and Structural Change

The starting point for this study is the economic-complexity literature, which models structural transformation through the joint distribution of specialization across locations and activities. In this tradition, revealed comparative advantage (RCA) is used to identify whether a location is meaningfully specialized in an activity, while diversity and ubiquity summarize, respectively, how many activities a location performs competitively and how widespread each activity is across locations [3,5,7]. Subsequent work also clarified the interpretation of the complexity indices themselves, showing that the ECI and PCI are linked to spectral partitioning and dimensionality reduction rather than to diversity alone [18]. The same literature introduced the product-space logic, according to which diversification is path dependent because new activities are more likely to emerge where related capabilities are already present [4]. In this view, structural change is not random movement over an unrestricted activity universe, but constrained expansion around an inherited portfolio of capabilities. A related contribution showed that the country–product network exhibits a systematic negative relationship between diversification and product ubiquity and that this pattern can be interpreted in terms of heterogeneous capability endowments across countries [19].

A second strand developed this logic beyond national export baskets and showed that relatedness matters for regional, firm-level and technological diversification as well. Regions are more likely to enter industries related to those already present locally [20], firm-level evidence likewise points to the role of related skill bases and industry-specific knowledge in diversification processes [21,22], and cities are more likely to develop technologies that are proximate to their incumbent knowledge bases [6,23]. This literature also

clarified that diversification is not only about feasibility. It also has a value dimension, because locations seek not merely to add activities, but to move toward more sophisticated ones [24]. In that sense, relatedness captures the compatibility of new activities with existing capabilities, whereas complexity captures their strategic value in terms of accumulated knowledge and competitive position [5,7].

The same debate also made clear that relatedness should not be treated as a single, self-evident empirical object. While the principle of relatedness is robust across many empirical domains, its operationalization depends on non-trivial choices about the productive unit in which co-occurrences are observed, the way proximity is calculated and the benchmark used to summarize the fit of a candidate activity to a local portfolio [17]. This point is especially important for studies that compare opportunities across heterogeneous locations. Raw relatedness density is mechanically influenced by the size and composition of the incumbent portfolio, so comparisons across locations are often better grounded when the signal is normalized relative to the relevant opportunity set rather than interpreted in absolute terms alone [7,17]. For the present article, this methodological issue affects how country–subfield opportunities are compared and therefore how predictive evidence is interpreted.

A further implication of this literature is the distinction between explanatory and predictive uses of relatedness. Several influential studies use relatedness primarily to explain observed diversification patterns ex post or to interpret path dependence substantively [4,6,20]. A predictive use, however, requires stronger discipline in model design because it must specify what exactly is being forecast, under which operationalization and with what benchmark of performance. The present study builds on that explanatory tradition but adopts a stricter predictive interpretation, treating variables derived from the topology as empirically assessable signals whose informational content must be examined before they are used in a prospective portfolio exercise.

2.2. Research-Space Approaches and Scholarly Diversification

The extension of relatedness ideas to scientific activity gave rise to research-space and knowledge-space approaches. Adjacent science-mapping work also showed that large-scale bibliographic data can be organized into global classification systems and overlay maps that represent the disciplinary structure of science and support the analysis of institutional and national profiles [25]. In the literature on technological knowledge, scholars mapped knowledge spaces by examining co-occurrence patterns in patents and other innovation data, thereby showing that non-trade domains also exhibit structured capability landscapes rather than unbounded variety [6,23,26]. Network-based approaches to science mapping have long shown that scientific fields can be studied as evolving relational structures, where changes in concepts, communities and their connections reveal the temporal organization of scientific activity [27]. In the scholarly domain, research-space studies further demonstrated that the emergence of new scientific activity is also path dependent. Guevara et al. showed that the probability of entering a new research field increases when that field is related to others in which the focal unit already has strong activity, and they documented this logic across individuals, institutions and countries [10]. Chinazzi et al. likewise showed that research domains can be organized as structured spaces whose topology is informative about future development patterns [11]. Related evidence at the individual level also shows that research-portfolio diversification is shaped by knowledge relatedness and social relatedness, reinforcing the view that scientific expansion is constrained by both cognitive proximity and collaboration structure [28].

These contributions establish three important points for the present study:

- Scientific activity can be represented using capability-space logic rather than only citation or topic-similarity logic.
- Scholarly diversification is sufficiently structured to support forward-looking opportunity assessment.
- Field evolution is contingent on changing relational structures, collaboration patterns and portfolio composition, so research-space signals should be interpreted as empirically assessed indicators rather than deterministic predictions.

Machine learning models provide a complementary way to study the temporal growth of academic fields from publication data. Recent work has used recurrent models, pretrained language models, graph neural networks and dynamic topic models to detect research trends, estimate technology popularity and trace changes in topics or technical terms over time. Du et al. [29], for example, propose a graph propagated neural topic model that combines citation relations, extracted technical terms and dynamic topic modeling to mine technology trends in scientific publications. Such approaches show that growth in academic fields can be examined through textual, citation and temporal signals, including signals that may not coincide fully with fixed field labels.

Despite these advances, most applications still focus on constructing or visualizing the space, forecasting field emergence, or modeling topic growth. These outputs are informative for understanding scientific change, but they do not generally translate into a portfolio problem in which candidate subfields are selected to meet an explicit target under a defined feasibility measure. This distinction matters because a space representation or a forecast of topic growth can be informative without providing a disciplined rule for portfolio choice.

Recent work also strengthened the relevance of scientific activity as part of a broader innovation system rather than as an isolated descriptive layer. Balland and Boschma showed that scientific capabilities in specific domains are a strong predictor of technological diversification in related domains, which implies that field-specific scientific strengths can provide meaningful information about future opportunity structures [30]. Similarly, Catalán et al. examined the scientific and technological cross-space and showed that endogenous scientific capabilities are relevant for understanding technological diversification opportunities [31]. Likewise, Hausmann et al. developed a multidomain innovation-complexity perspective in which scientific publications, patents and trade capture related but distinct layers of capability accumulation, and they showed that diversification opportunities can also be inferred across these domains [9]. These studies are highly relevant for scholarly diversification because they place scientific capabilities within a larger innovation architecture and therefore strengthen the policy relevance of understanding how research portfolios evolve.

However, neither stream fully resolves the specific problem addressed here. Research-space studies in science typically stop at descriptive topology or predictive field-emergence analysis [10,11]. Multidomain innovation-complexity studies clarify why scientific capabilities matter, but they do not formulate a country–subfield portfolio-design framework that is internal to the scholarly domain itself [9]. In addition, OpenAlex-based bibliometric work has shown that open data can support meaningful science mapping at individual, institutional and national levels [13], but mapping capability alone does not amount to a diversification framework. It does not specify how to transform a scholarly activity space into a historically assessed predictive model and then into a target-oriented decision rule. This limitation motivates the empirical integration developed in the present study.

2.3. From Relatedness Heuristics to Policy-Oriented Prioritization and Optimization

A separate literature has asked how relatedness and complexity can inform policy prioritization. Within smart-specialization debates, relatedness is usually interpreted as a proxy for feasibility and complexity as a proxy for strategic value, making the desirable target a domain that is both sufficiently connected to incumbent capabilities and sufficiently sophisticated to support upgrading [32]. This logic has been highly influential, but the empirical record is more mixed than policy narratives sometimes suggest. Deegan et al. show that actual priority choices often incorporate relatedness and complexity independently rather than jointly, and that regional strategies may still fall short of the place-based differentiation envisioned in theory [32]. This concern is reinforced by evidence that smart-specialization priorities should be assessed against the structure of the local knowledge space, rather than only against broad strategic labels or aspirational domains [33]. Building on this policy-oriented use of relatedness and complexity, recent contributions have proposed ranking procedures that combine several indicators in order to identify promising diversification targets, often distinguishing between shorter- and longer-term opportunities [34]. These contributions are policy-relevant because they move beyond pure description, but they still largely take the form of ranking systems.

Relatedness–complexity diagrams (popularized in applied diversification analysis by The Atlas of Economic Complexity [8]) provided an influential visual heuristic for identifying activities that appear both feasible and strategically valuable. The limitation of ranking-based prioritization is that it usually presumes that candidates can be ordered one by one according to a scalar score and then selected sequentially. Once the policy problem is framed as attainment of a target under constraints, the best portfolio need not coincide with the top of a relatedness ranking or with the set of activities located in a visually attractive region of a relatedness-complexity diagram. For instance, Alshamsi et al. formalized diversification as a strategic diffusion problem and showed that optimal strategies need not correspond to greedily choosing the most connected next activity at every step [15]. More directly, Stojkoski and Hidalgo argue that relatedness-complexity diagrams are useful heuristics but do not embody an explicit optimality criterion, because they rely on proxy measures and omit higher-order effects, future shifts in complexity and explicit portfolio constraints [16]. Their response is to introduce a target-oriented optimization framework that minimizes a cost function subject to a formal diversification objective.

More broadly, metaheuristic algorithms are often used in complex system design and parameter-estimation settings when exhaustive search is impractical or the search process requires flexible exploration strategies, as illustrated by recent work on PEM fuel-cell parameter extraction [35]. In the present study, after the eligible set and the complexity target are defined, the portfolio layer can be expressed as a binary linear optimization problem with an explicit objective and a linear target-attainment constraint. We therefore use a mixed-integer linear programming formulation for the reported instances, which is well aligned with the structure of the problem and allows the comparison with the relatedness-ranking benchmark to be made under the same target condition.

This optimization turn is particularly important for the positioning of the present study, because it raises the standard against which a scholarly diversification study should be evaluated. Mapping a research space or ranking candidate subfields by relatedness may be descriptively useful, but remains vulnerable to the criticism that it has not demonstrated a sufficiently clear methodological advance beyond existing approaches. Building on recent work that has connected relatedness-based diversification analysis with optimization-oriented reasoning, we aim to make a more specific methodological contribution by showing empirically how historical signals from a scholarly research space are used in an explicit target-oriented portfolio problem.

3. Methods

This section presents our empirical framework for scholarly diversification across countries and subfields, summarized schematically in Figure 1. The framework is organized in two coupled layers:

- The first layer constructs a research space under a common protocol and evaluates whether measures derived from that space help predict subsequent structural change.
- The second layer uses the signals retained from the historical analysis in a prospective portfolio problem.

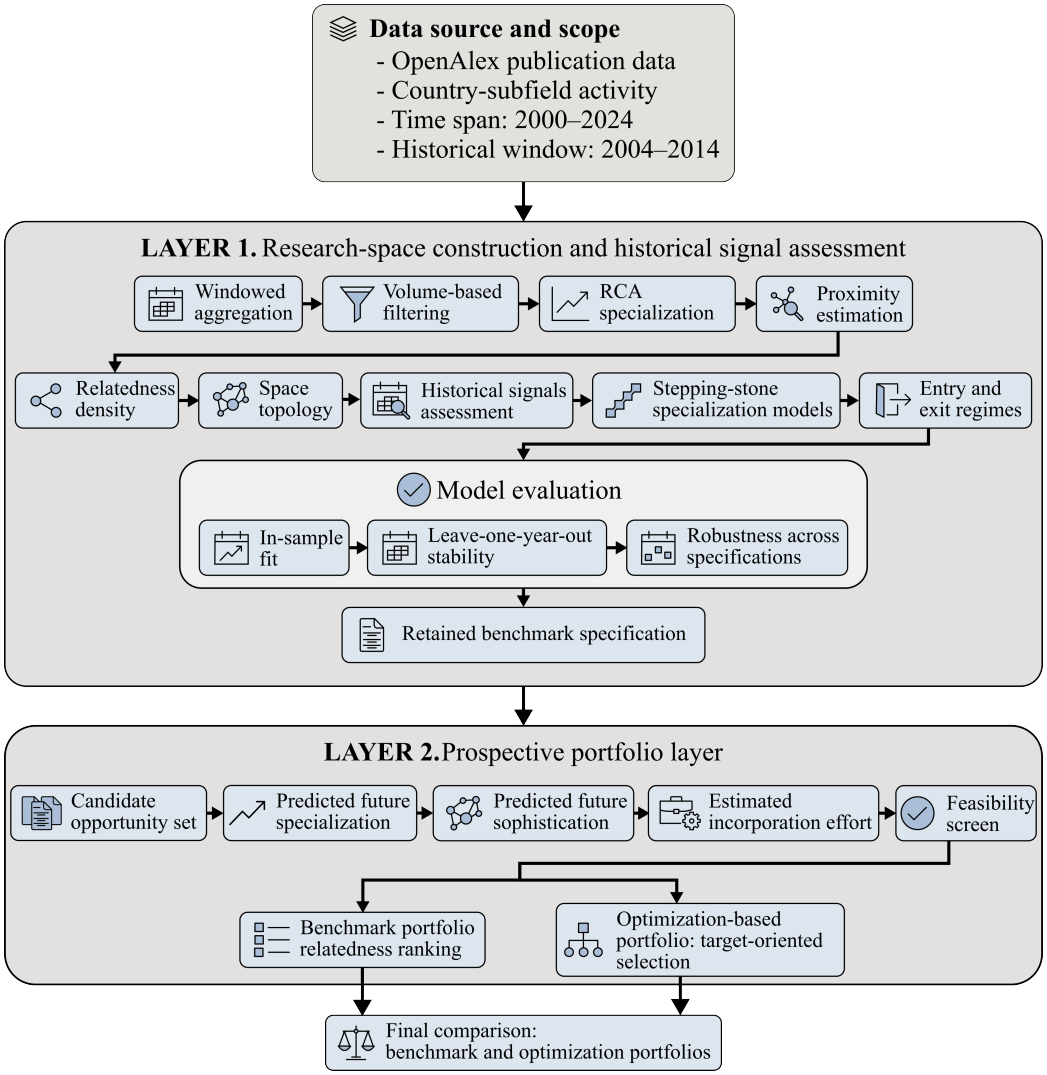


Figure 1. Schematic overview of the empirical framework for country–subfield scholarly diversification.

We first describe the data source, temporal coverage, aggregation levels and analytical scope. We then explain how annual country–subfield activity is transformed into a research-space topology through windowed aggregation, specialization, proximity estimation and relatedness density. Next, we introduce the framework for historical assessment used to estimate and evaluate a multivariate stepping-stone model of future specialization. Finally, we define the portfolio layer used to compare the benchmark portfolios with those selected by optimization for recent country–year cases.

3.1. Data Source and Empirical Scope

The empirical analysis is based on OpenAlex [12], a free and open index of global scholarly research that provides large-scale metadata on scientific outputs and their associated

entities. OpenAlex is well suited to the present analysis because it offers broad international coverage and a standardized representation of the scholarly record, which makes it possible to measure research activity across countries and subfields in a consistent and reproducible way. This choice is aligned with recent work showing that OpenAlex can support meaningful bibliometric mapping at the individual, institutional and national levels [13] and with recent multidomain complexity analyses that use OpenAlex publication data jointly with patent and trade data to characterize scientific, technological and industrial capabilities at the country level [9].

We collected data for the period 2000–2024 and organized the records into annual document counts at three aggregation levels (in the aggregation, the country dimension represents the country associated with the affiliation metadata of publications indexed in OpenAlex. The subfield dimension refers to the OpenAlex subfield-level subject category used as the analytical unit of scholarly activity. The year dimension is the calendar year of publication):

- Country–subfield–year. For each country c, subfield s and year y, the data record the corresponding annual publication activity, later denoted by $x_{c,s,y}$. This level is used to construct RCA, binary specialization indicators, proximity, relatedness density, entry–exit events and portfolio candidates.
- Country–year. This marginal level aggregates publication activity over subfields. It is used to compute country totals and country-level volume filters.
- Subfield–year. This marginal level aggregates publication activity over countries. It is used to compute subfield totals and subfield-level volume filters.

Together, these records provide the country–subfield activity matrix and the country and subfield marginals required for RCA normalization, volume filtering and research-space construction. This keeps the data structure aligned with the proximity and relatedness measures introduced below.

Although the underlying extraction window spans 2000–2024, the benchmark analyses reported in this paper focus on the 2004–2014 evaluation period. This restriction follows from the temporal logic of the design, as research-space construction relies on backward-looking aggregation windows, while structural change is evaluated over a forward-looking horizon. The broader OpenAlex retrieval window therefore provides the temporal support needed to compute rolling activity matrices and subsequent transitions without edge truncation, whereas the 2004–2014 benchmark window defines the observations on which we assess the topology and its predictive content.

3.2. Research-Space Construction

Starting from the country–subfield–year activity data described above, we construct the annual activity matrices and transform them into a research-space topology in four steps following the standard logic of economic complexity:

1. Aggregate annual country–subfield activity into windowed matrices.
2. Derive a binary specialization signal from revealed comparative advantage.
3. Estimate proximity between subfields from patterns of co-specialization.
4. Summarize the position of each country–subfield pair through relatedness density.

In the broader literature, the same country–activity quantity is sometimes referred to as *relatedness density* and, in some research-space applications, as *knowledge density* [4,7,10,11]. To remain aligned with the core economic-complexity terminology, we use *proximity* for the activity–activity relation and *relatedness density* for the country–subfield relation.

3.2.1. Windowed Activity Matrix and Volume-Based Filtering

Let $x_{c,s,y} \geq 0$ denote the annual document count of country c in subfield s in calendar year y. For a target year t and a backward-looking aggregation window of length h, we define the aggregated country–subfield matrix as

$$X_{c,s}^{(t,h)} = \sum_{y=t-h+1}^{t} x_{c,s,y}, \tag{1}$$

with country totals $X_{c,\cdot}^{(t,h)} = \sum_s X_{c,s}^{(t,h)}$, subfield totals $X_{\cdot,s}^{(t,h)} = \sum_c X_{c,s}^{(t,h)}$, and grand total $X_{\cdot,\cdot}^{(t,h)} = \sum_{c,s} X_{c,s}^{(t,h)}$.

To improve numerical stability and reduce the influence of extremely sparse entities and cells, we filter by volume using a cutoff for country totals, thr_C; a cutoff for subfield totals, thr_S; and a cutoff for individual cells, thr_{CS}. The retained sets are

$$\mathcal{C}^{(t,h)} = \left\{ c : X_{c,\cdot}^{(t,h)} > thr_C \right\}, \qquad \mathcal{S}^{(t,h)} = \left\{ s : X_{\cdot,s}^{(t,h)} > thr_S \right\}, \tag{2}$$

and we keep only cells satisfying $X_{c,s}^{(t,h)} > thr_{CS}$ for $(c,s) \in \mathcal{C}^{(t,h)} \times \mathcal{S}^{(t,h)}$.

Table 1 summarizes the baseline configuration, which separates the aggregation window used in the empirical analyses from the longer window used solely for visualization. All reported estimation, validation and robustness results are computed under the baseline inference window. The longer window is introduced only to stabilize the graphical layout of the network and does not enter the empirical pipeline used for the reported results.

Table 1. Baseline configuration used to construct the research space and to run the predictive-validation experiments. The longer window is used only to stabilize the graphical layout, whereas all empirical analyses use the baseline inference window.

Component	Choice
Inference window (baseline)	$h = 5$ years
Visualization window (layout only)	$h = 10$ years
Filtering mode	volume-filtered sample
Country total threshold	$thr_C = 1250$
Subfield total threshold	$thr_S = 250$
Cell threshold	$thr_{CS} = 50$
Minimum ubiquity	$ub = 3$
Minimum diversity	$dv = 3$
Proximity symmetrization	minimum conditional probability

To quantify the information preserved after filtering, we report the retained mass,

$$\text{Retained_Mass}^{(t,h)} = \frac{\sum_{c \in \mathcal{C}^{(t,h)}} \sum_{s \in \mathcal{S}^{(t,h)}} X_{c,s}^{(t,h)} \mathbf{1}\left[X_{c,s}^{(t,h)} > thr_{CS} \right]}{\sum_c \sum_s X_{c,s}^{(t,h)}}, \tag{3}$$

where $\mathbf{1}[\cdot]$ denotes the indicator function, equal to 1 when the condition in brackets is satisfied and 0 otherwise, so that only retained cells contribute to the numerator.

3.2.2. Specialization, Proximity and Relatedness Density

Once the activity matrix has been filtered, we define specialization using the standard revealed comparative advantage criterion introduced by Balassa [3] and widely used throughout the economic-complexity literature [4,5,7]. Equations (4) and (5) follow this

RCA-threshold convention, where RCA ≥ 1 identifies country–subfield specialization. We first compute

$$\mathrm{RCA}_{c,s}^{(t,h)} = \frac{X_{c,s}^{(t,h)} / X_{c,\cdot}^{(t,h)}}{X_{\cdot,s}^{(t,h)} / X_{\cdot,\cdot}^{(t,h)}}, \tag{4}$$

and define the corresponding binary specialization indicator,

$$M_{c,s}^{(t,h)} = \mathbf{1}\Big[\mathrm{RCA}_{c,s}^{(t,h)} \geq 1\Big]. \tag{5}$$

Consistent with the standard diversity–ubiquity notation used in economic-complexity analysis [5,7], filtered diversity and ubiquity are then

$$k_{c,0}^{(t,h)} = \sum_{s} M_{c,s}^{(t,h)}, \qquad k_{s,0}^{(t,h)} = \sum_{c} M_{c,s}^{(t,h)}, \tag{6}$$

and we retain only countries with $k_{c,0}^{(t,h)} \geq dv$ and subfields with $k_{s,0}^{(t,h)} \geq ub$.

To construct the research space, we estimate proximity between subfields from patterns of co-specialization. Following the canonical product-space and technology-space tradition, Equations (7) and (8) define proximity as the minimum of the pairwise conditional probabilities of specialization [2,4,6]. Let

$$P^{(t,h)}(s \mid s') = \Pr\Big(M_{c,s}^{(t,h)} = 1 \,\Big|\, M_{c,s'}^{(t,h)} = 1\Big) = \frac{\sum_c M_{c,s}^{(t,h)} M_{c,s'}^{(t,h)}}{\sum_c M_{c,s'}^{(t,h)}}, \tag{7}$$

and analogously $P^{(t,h)}(s' \mid s)$. We then define subfield proximity as

$$\phi_{s,s'}^{(t,h)} = \min\Big\{P^{(t,h)}(s \mid s'),\, P^{(t,h)}(s' \mid s)\Big\}. \tag{8}$$

In the empirical implementation, Equation (8) is operationalized through the OEC-style `procedure=max` option. Under binary specialization, this implementation is equivalent to computing co-specialization intensity relative to the larger of the two marginal specialization counts, which yields the same symmetric proximity based on the minimum conditional probability used as the benchmark definition.

This choice aligns the manuscript with the standard proximity convention used in relatedness-based activity-space applications across empirical domains, where symmetric affinity is derived from co-specialization patterns through the minimum of the two conditional probabilities [4,6]. It differs from the maximum conditional probability used in some research-space visualizations based on asymmetric author-sharing links, where the goal is primarily graphical simplification rather than the construction of a canonical symmetric relatedness measure [10]. Because the present study is formulated as a country–activity co-specialization problem, the minimum conditional probability provides a more appropriate baseline.

Using the proximity matrix, and following the standard relatedness-density formulation used in product-space and related-diversification studies [4,6–8], we summarize the position of a country–subfield pair through relatedness density,

$$\omega_{c,s,t} = \frac{\sum_{s' \neq s} M_{c,s',t}\, \phi_{s,s'}}{\sum_{s' \neq s} \phi_{s,s'}}, \tag{9}$$

which is the weighted share of proximate activities already present in the country's portfolio. By construction, $\omega_{c,s,t} \in [0,1]$. A high value indicates that the candidate subfield is surrounded by many proximate activities in which the country is already specialized.

Because relatedness density is mechanically correlated with portfolio size, comparisons across countries or subfields are better made using a *relative relatedness* measure and

an explicit benchmark support set [7,17]. We therefore define the benchmark relatedness term as the density standardized over the country-specific opportunity set,

$$\widetilde{\omega}^{\mathcal{O}}_{c,s,t} = \frac{\omega_{c,s,t} - \mu^{\mathcal{O}}_{c,t}}{\sigma^{\mathcal{O}}_{c,t}}, \qquad \mathcal{O}_{c,t} = \{s : \mathrm{RCA}_{c,s,t} < 1\}, \tag{10}$$

where $\mu^{\mathcal{O}}_{c,t}$ and $\sigma^{\mathcal{O}}_{c,t}$ are the mean and standard deviation of density over the contemporaneous opportunity set of country c at time t. In the notation used elsewhere in the manuscript, the label *z-score + opportunities* denotes relative relatedness standardized over the opportunity set. This specification preserves the core density logic while normalizing for cross-country differences in opportunity-set composition [7].

The resulting weighted network on the retained subfields constitutes the *research space*. The term is used here in the same broad sense as in prior work that maps research fields through shared productive capabilities rather than through direct citation flows [10,11]. The construction of the network is fully determined by Equations (1)–(10); any subsequent drawing is a visualization choice and does not alter the computed proximities or density values.

3.2.3. Baseline Sample Characteristics and Visual Layout

Under the baseline configuration, the retained sample remains large throughout the analytical period and preserves most of the total research mass. This is important because the predictive stage is meant to assess the empirical content of the topology rather than to exploit an artificially narrow subset of the data. The year-by-year scale of the retained system is summarized in Table 2, which reports the number of retained countries, retained subfields, active country–subfield cells, and the retained mass defined in Equation (3) for each year from 2004 to 2014. The restriction from the raw universe to the retained filtered sample is reported separately in Appendix A Table A1.

Table 2. Baseline thresholds and sample sizes across 2004–2014 ($h = 5$; $thr_C = 1250$, $thr_S = 250$, $thr_{CS} = 50$). Retained mass is defined in Equation (3).

Year	$n_{\text{countries}}$	$n_{\text{subfields}}$	n_{cells}	Retained_Mass
2004	101	248	8730	0.9782
2005	102	249	9109	0.9797
2006	105	249	9535	0.9811
2007	108	249	9975	0.9821
2008	114	250	10,434	0.9834
2009	121	250	10,967	0.9846
2010	124	250	11,500	0.9856
2011	126	250	11,960	0.9861
2012	131	251	12,518	0.9867
2013	137	251	13,108	0.9871
2014	142	251	13,734	0.9876

The volume filter of the baseline configuration reported in Table 1 is also applied to the 2024 country–subfield activity matrix using the baseline inference window ($h = 5$). This choice keeps the prospective sample construction aligned with the historical assessment, where the same filtering protocol preserves an average retained mass of 0.9838 over the 2004–2014 benchmark period. Applying the protocol to 2024 retains 164 countries and a retained mass of 0.9914 for the prospective cross-country comparison. Appendix A Table A1 extends the sample-restriction and retained-mass summary through 2024.

For descriptive orientation, Figure 2 provides a visualization of the retained research-space topology, allowing to inspect its broad structural organization, the relative positioning

of subfields, and the contrast between denser and more peripheral regions of the network. To make these large-scale features easier to perceive, the figure is drawn using a longer aggregation window ($h = 10$ years) with the same filtering thresholds and structural rules as the baseline. This choice stabilizes the graphical layout across adjacent years without changing the empirical design used for inference. All historical assessment results reported in Section 4 use the baseline inference window $h = 5$. The research-space visualization in Figure 2 was prepared using Gephi 0.11.2.

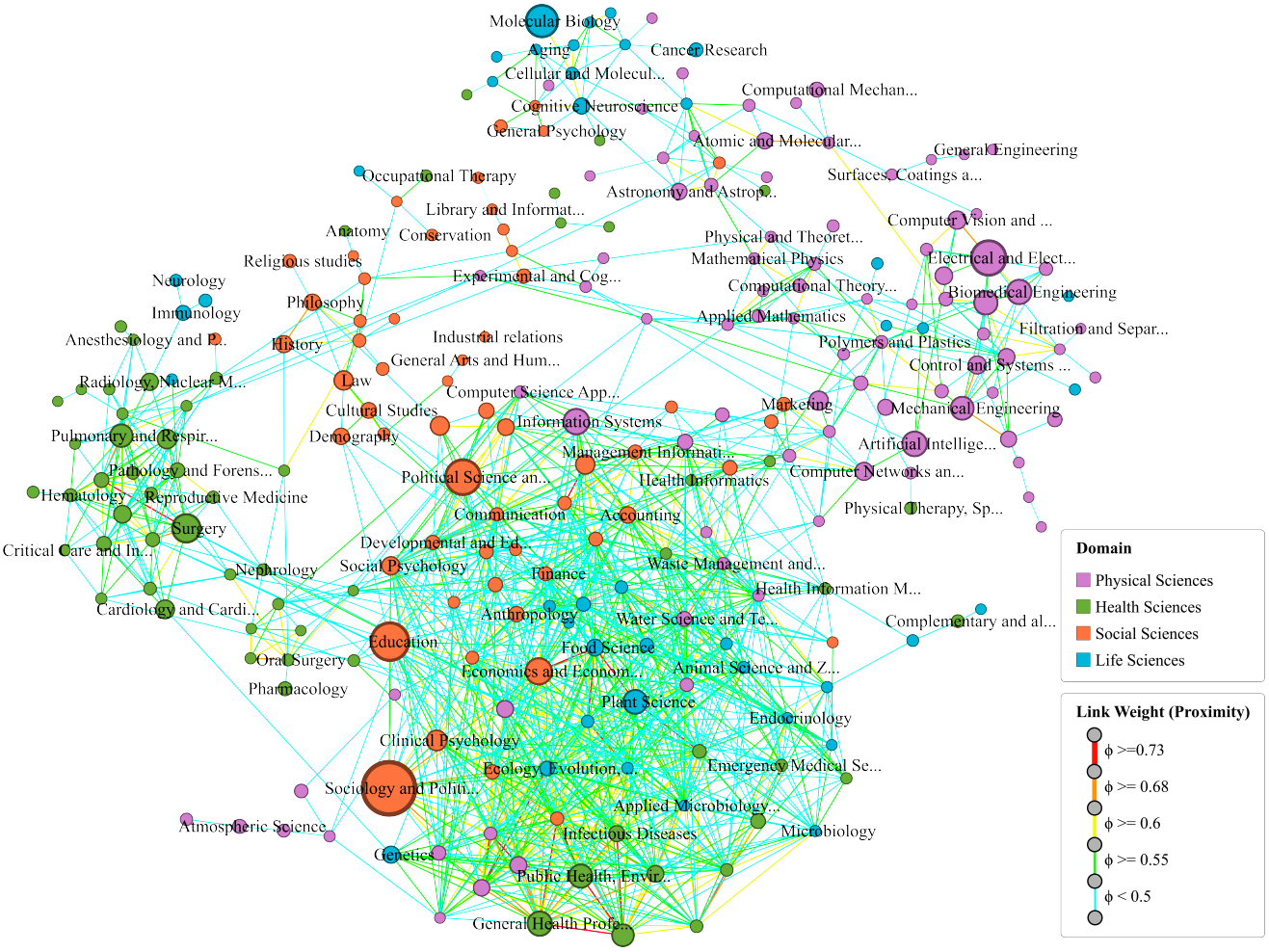


Figure 2. Research-space visualization built with a 10-year window ($h = 10$) for layout stability and the baseline filtering thresholds ($thr_C = 1250$, $thr_S = 250$, $thr_{CS} = 50$), together with $ub = 3$, $dv = 3$, and minimum-conditional-probability proximity.

3.3. Historical Assessment Framework

Once the research space has been constructed, the first analytical layer evaluates whether the resulting topology-derived signals contain empirical information about subsequent structural change. The historical assessment is model-based. If the research space captures meaningful relations among subfields, then a country's position near a candidate subfield should help explain later entry into specialization, and exit from specialization, beyond persistence alone. To examine this proposition, we define regime-specific transition outcomes, estimate reduced-form stepping-stone equations for future transformed specialization, and assess their performance using both in-sample fit and a leave-one-year-out stability check. We then compare nested specifications to isolate the marginal contribution of the topology-derived terms, especially absolute relatedness density and relative related-

ness standardized over the opportunity set. This first layer tests whether the signals later used in the forward-looking design stage have stable predictive value.

Consistent with the temporal structure used in this study, the benchmark setting combines a base-year observation at t, an intermediate stepping-stone observation at $t+\tau$, and an evaluation horizon at $t+\Delta t$. Throughout the benchmark analyses we use $\tau = 5$ and $\Delta t = 10$ under the baseline construction protocol. The resulting model is therefore not a pure single-origin forecast from t alone, but a reduced-form stepping-stone specification that assesses whether current topology-derived signals at t, together with intermediate specialization observed at $t+\tau$, help account for specialization intensity at $t+\Delta t$.

Let c index countries, s index subfields, and t denote the observation year. We work with the transformed specialization measure

$$r_{c,s,t} \equiv \log(1 + \mathrm{RCA}_{c,s,t}), \tag{11}$$

and define the intermediate stepping-stone specialization term as

$$r_{c,s,\tau} \equiv r_{c,s,t+\tau}. \tag{12}$$

Equation (12) refers to the historical assessment sample, where the intermediate stepping-stone observation at $t+\tau$ is available by construction. Under the benchmark setting used in this paper ($\tau = 5$, $\Delta t = 10$), this condition is satisfied for the 2004–2014 base years included in the validation exercises. Recent prospective cases such as base year 2024 are therefore outside the observed historical validation sample based on this three-period structure and are treated separately in the portfolio layer through an explicit prospective stepping-stone convention introduced in Section 3.4. This distinction is important because the historical layer evaluates the model with observed information at $t+\tau$, whereas the prospective layer applies the historically calibrated specification under the information actually available at the decision date.

Following the threshold-crossing logic used in related-diversification studies [4,6,7,20], future structural change over horizon Δt is summarized through the standard entry and exit events

$$E_{c,s,t}^{(\Delta t)} = \mathbf{1}[\mathrm{RCA}_{c,s,t} < 1 \ \wedge \ \mathrm{RCA}_{c,s,t+\Delta t} \geq 1], \tag{13}$$

$$X_{c,s,t}^{(\Delta t)} = \mathbf{1}[\mathrm{RCA}_{c,s,t} \geq 1 \ \wedge \ \mathrm{RCA}_{c,s,t+\Delta t} < 1]. \tag{14}$$

Henceforth, the entry regime refers to observations in the contemporaneous opportunity set $\mathcal{O}_{c,t}$, whereas the exit regime refers to incumbent specialized activities.

We estimate regime-specific reduced-form transition equations for future transformed specialization, separately for the entry and exit regimes:

$$r_{c,s,t+\Delta t} = \alpha^{(g)} + \beta_1^{(g)} r_{c,s,\tau} + \beta_2^{(g)} r_{c,s,t} + \beta_3^{(g)} \omega_{c,s,t} + \beta_4^{(g)} \widetilde{\omega}_{c,s,t}^{\mathcal{O}} + \varepsilon_{c,s,t}^{(g)}, \qquad g \in \{E, X\} \tag{15}$$

This transition specification is formulated in terms of future transformed specialization rather than as a binary response model for threshold crossing. This choice aligns with the purpose of assessing whether the benchmark representation of the research space carries systematic information about subsequent changes in specialization intensity within the entry and exit regimes. The threshold-crossing definitions in Equations (13) and (14) remain relevant, however, because they provide the event-based interpretation used for the complementary ranking diagnostics reported below.

Consistent with this reduced-form perspective, our main goodness-of-fit criterion is the regime-specific continuous fit, R^2_{cont}. Since the fitted values from Equation (15) also induce an ordering over future entry and exit events, we additionally report AUC as a complementary discrimination measure. Other diagnostics, including precision, recall, F_1,

PR-AUC, pseudo-Brier scores, placebo benchmarks and reliability checks, are treated as ancillary evidence and are therefore reported in Appendix C.

The year-holdout exercise reported in the main text is leave-one-year-out (LOYO) over base years. In implementation, the training set excludes the held-out base year but may still contain years later than the held-out observation. In this setting, LOYO functions as a stability and transportability check across annual slices. Its role in this paper is to assess the temporal robustness of the signal structure in a year-holdout setting.

To assess the incremental contribution of the topology-derived terms, we estimate a nested sequence of specifications. The baseline persistence model includes only lagged specialization, $\mathcal{S}_1 = \{r_{c,s,\tau}\}$. We then add contemporaneous specialization, $\mathcal{S}_2 = \{r_{c,s,\tau}, r_{c,s,t}\}$, followed by absolute relatedness density, $\mathcal{S}_3 = \{r_{c,s,\tau}, r_{c,s,t}, \omega_{c,s,t}\}$. The full specification augments this set with opportunity-set standardized relative relatedness, $\mathcal{S}_4 = \{r_{c,s,\tau}, r_{c,s,t}, \omega_{c,s,t}, \widetilde{\omega}^{\mathcal{O}}_{c,s,t}\}$. This nesting structure makes it possible to determine whether the explanatory gain comes primarily from persistence, from absolute topological position or from the relative opportunity signal captured by $\widetilde{\omega}^{\mathcal{O}}_{c,s,t}$.

3.4. Portfolio Design with an Explicit Target

The second layer of our framework formulates the prospective diversification exercise as a portfolio problem with an explicit target. This design is conceptually aligned with prior work on optimal diversification in networks of related products and research areas [15] and with the target-oriented optimization logic developed by Stojkoski and Hidalgo [16]. These contributions show that relatedness-based heuristics and formal optimization criteria need not lead to the same diversification choices. The present study adapts this logic to the scholarly domain. It does so within an empirical framework in which the representation of the research space used for portfolio design is first assessed historically under a common construction protocol and then used in the prospective stage.

The portfolio layer is applied to country–year cases retained from the filtered country–subfield system. For each retained country c and base year t, specialized subfields define the incumbent portfolio, while $\mathcal{O}_{c,t}$ denotes the corresponding opportunity set of non-specialized subfields. After the feasibility screen described below, the remaining candidates form the final eligible set $\mathcal{E}_{c,t}$ for that country. This same eligible set is used for both the relatedness-ranking benchmark and the optimized portfolio.

For each retained country c and base year t in the portfolio layer, let

$$\mathcal{S}_{c,t} = \{s : \mathrm{RCA}_{c,s,t} \geq 1\} \tag{16}$$

denote the incumbent specialized portfolio. The initial candidate pool is the country-specific opportunity set $\mathcal{O}_{c,t}$, consisting of subfields in which country c is not yet specialized, as defined in Equation (10). The portfolio problem uses two quantities from the benchmark stepping-stone model, namely predicted future sophistication and estimated entry effort:

- We first compute predicted future transformed specialization for every country–subfield pair by inserting the observed base-year covariates into the benchmark transition equation from Equation (15). However, for recent prospective base years, the intermediate stepping-stone observation at $t + \tau$ is not yet observed. We therefore define the term entering the forward-looking application as

$$r^{\dagger}_{c,s,\tau} = \begin{cases} r_{c,s,t+\tau}, & \text{if the intermediate observation is available,} \\ r_{c,s,t}, & \text{otherwise.} \end{cases} \tag{17}$$

  For recent prospective cases such as base year 2024, the second branch applies because the matrix at $t + \tau$ is not yet observed. This carry-forward convention preserves the

contemporaneous specialization position as the stepping-stone baseline and avoids imposing an artificial zero intermediate state. Using this definition, predicted future transformed specialization is

$$\widehat{r}_{c,s,t+\Delta t} = \widehat{\alpha}^{(g)} + \widehat{\beta}_1^{(g)} r^{\dagger}_{c,s,\tau} + \widehat{\beta}_2^{(g)} r_{c,s,t} + \widehat{\beta}_3^{(g)} \omega_{c,s,t} + \widehat{\beta}_4^{(g)} \widetilde{\omega}^{\mathcal{O}}_{c,s,t}, \qquad g \in \{E, X\}, \quad (18)$$

which yields the implied future specialization ratio

$$\widehat{\mathrm{RCA}}_{c,s,t+\Delta t} = \exp(\widehat{r}_{c,s,t+\Delta t}) - 1. \quad (19)$$

Using the conventional specialization threshold, we define the predicted presence matrix

$$\widehat{M}_{c,s,t+\Delta t} = \mathbf{1}\Big[\widehat{\mathrm{RCA}}_{c,s,t+\Delta t} \geq 1\Big]. \quad (20)$$

Applying the same subfield-complexity operator used in the empirical pipeline, following the economic-complexity tradition on activity complexity [5,7,18], to the full cross-country predicted matrix $\widehat{M}_{t+\Delta t}$ yields the corresponding future subfield sophistication scores,

$$\widehat{\mathrm{PCI}}_{s,t+\Delta t} = \mathcal{C}_s\Big(\widehat{M}_{t+\Delta t}\Big), \quad (21)$$

where $\mathcal{C}_s(\cdot)$ denotes the PCI mapping induced by the complexity algorithm on the predicted specialization matrix. Operationally, future PCI is not forecast separately for each candidate subfield. Rather, it is recomputed jointly from the full cross-country predicted specialization matrix $\widehat{M}_{t+\Delta t}$, so that the resulting sophistication scores remain internally consistent with the complexity framework. These quantities provide the forward-looking sophistication terms used in the portfolio-design stage.

- Secondly, we obtain an estimated entry-effort term. For entry candidates, the benchmark implementation inverts the entry equation to compute the transformed specialization level required to attain a target specialization threshold $\mathrm{RCA}^{\star}$, with $\mathrm{RCA}^{\star} = 1$ in the baseline design. Specifically,

$$\widehat{r}^{\star}_{c,s,\tau} = \frac{\log(1 + \mathrm{RCA}^{\star}) - \left[\widehat{\alpha}^{(E)} + \widehat{\beta}_2^{(E)} r_{c,s,t} + \widehat{\beta}_3^{(E)} \omega_{c,s,t} + \widehat{\beta}_4^{(E)} \widetilde{\omega}^{\mathcal{O}}_{c,s,t}\right]}{\widehat{\beta}_1^{(E)}}, \quad (22)$$

and the corresponding candidate-specific effort proxy is

$$\widehat{Y}^{\star}_{c,s,t} = \max\{0, \exp(\widehat{r}^{\star}_{c,s,\tau}) - 1 - \mathrm{RCA}_{c,s,t}\}. \quad (23)$$

This quantity is interpreted as the additional stepping-stone specialization required for candidate s to become specialized at the target threshold under the benchmark transition model.

For recent prospective base years in which the intermediate observation at $t + \tau$ is unavailable, Equation (22) is evaluated under the same convention introduced in Equation (17). That is, the required stepping-stone term is evaluated relative to the contemporaneous specialization baseline at t, rather than relative to an unobserved intermediate matrix or an artificially imposed zero state.

In the portfolio layer, and following the target-attainment perspective of optimization-based economic-complexity analysis [16], the target is not the country-level ECI score itself. Rather, we use a portfolio-complexity proxy defined as the average PCI of the subfields in which the country is already specialized. We therefore write

$$\kappa^{\mathrm{raw}}_{c,t} = \frac{1}{|\mathcal{S}_{c,t}|} \sum_{u \in \mathcal{S}_{c,t}} \mathrm{PCI}^{\mathrm{use}}_{u,t}, \quad (24)$$

and define the target level as

$$\kappa^{\star}_{c,t} = \kappa^{\mathrm{raw}}_{c,t} + \Delta\kappa_{\mathrm{raw}}. \tag{25}$$

The increment $\Delta\kappa_{\mathrm{raw}}$ specifies the target increase in raw portfolio complexity. Higher values impose a stricter target-attainment condition and may therefore change the number and composition of candidate subfields required to satisfy the portfolio constraint. Comparing portfolios across alternative values of $\Delta\kappa_{\mathrm{raw}}$ provides a direct way to examine how the target level affects portfolio size, selected candidates and predicted future sophistication.

Adapting the target-oriented optimization logic of Stojkoski and Hidalgo [16] to the country–subfield setting, we define binary decision variables $z_{c,s} \in \{0,1\}$ for $s \in \mathcal{E}_{c,t}$ and solve the minimum-effort portfolio problem

$$\min_{\{z_{c,s}\}} \sum_{s \in \mathcal{E}_{c,t}} \widehat{Y}^{\star}_{c,s,t}\, z_{c,s} \tag{26}$$

subject to the target-attainment condition

$$\frac{\sum_{u \in \mathcal{S}_{c,t}} \mathrm{PCI}^{\mathrm{use}}_{u,t+\Delta t} + \sum_{s \in \mathcal{E}_{c,t}} \mathrm{PCI}^{\mathrm{use}}_{s,t+\Delta t}\, z_{c,s}}{|\mathcal{S}_{c,t}| + \sum_{s \in \mathcal{E}_{c,t}} z_{c,s}} \geq \kappa^{\star}_{c,t}. \tag{27}$$

The empirical implementation distinguishes two effort-related quantities:

- The portfolio problem in Equation (26) is solved using the candidate-specific cost proxy $\widehat{Y}^{\star}_{c,s,t}$.
- For visualization and comparative plotting, we retain the auxiliary volume proxy $W^{cp}_{c,s,t}$ exported by the predictive pipeline. This quantity is used only as the horizontal coordinate in the portfolio scatter plots and does not enter the optimization objective.

The portfolio equations also use a single sophistication term, defined as

$$\mathrm{PCI}^{\mathrm{use}}_{s,t+\Delta t} = \begin{cases} \widehat{\mathrm{PCI}}_{s,t+\Delta t}, & \text{if a predicted future PCI is available,} \\ \mathrm{PCI}_{s,t}, & \text{otherwise.} \end{cases} \tag{28}$$

The final eligible set $\mathcal{E}_{c,t}$ is obtained by applying a feasibility screen on predicted sophistication to the opportunity set $\mathcal{O}_{c,t}$. Specifically, candidate subfields are retained only if their forward-looking sophistication remains within an ECI-conditioned admissible range. To construct that range, countries are grouped in the base year into bands of similar current ECI, and the distribution of observed specialized-subfield PCI within each band is summarized by its upper quartile and interquartile range. Thus, the raw admissible sophistication cap for band b is defined as

$$\overline{\mathrm{PCI}}^{\mathrm{cap}}_{b,t} = Q_{3,b,t} + IQR_{b,t}, \tag{29}$$

where $Q_{3,b,t}$ and $IQR_{b,t}$ denote, respectively, the third quartile and interquartile range of PCI among specialized subfields observed for countries in ECI band b at year t.

To avoid discontinuities across adjacent ECI bands, these band-level caps are smoothed as a function of band-midpoint ECI and then evaluated at the focal country's current ECI, yielding an admissible sophistication ceiling, $\overline{\mathrm{PCI}}^{\mathrm{cap}}_{c,t}$, for each country. Accordingly, the final eligible set used in the optimization and benchmark comparison is

$$\mathcal{E}_{c,t} = \left\{ s \in \mathcal{O}_{c,t} : \mathrm{PCI}^{\mathrm{use}}_{s,t+\Delta t} \leq \overline{\mathrm{PCI}}^{\mathrm{cap}}_{c,t} \right\}. \tag{30}$$

This restriction is intended to prevent the portfolio-design exercise from selecting candidate subfields whose predicted sophistication lies implausibly far above the range empirically associated with countries at comparable current complexity levels. The incumbent portfolio

$\mathcal{S}_{c,t}$ provides the starting set of specialized subfields, while the final eligible set $\mathcal{E}_{c,t}$ defines the candidate additions.

For computation, Equation (27) is implemented in its equivalent linear form,

$$\sum_{s \in \mathcal{E}_{c,t}} \left(\mathrm{PCI}^{\mathrm{use}}_{s,t+\Delta t} - \kappa^{\star}_{c,t}\right) z_{c,s} \geq \kappa^{\star}_{c,t} |\mathcal{S}_{c,t}| - \sum_{u \in \mathcal{S}_{c,t}} \mathrm{PCI}^{\mathrm{use}}_{u,t+\Delta t}. \tag{31}$$

The resulting optimization problem is therefore implemented as a binary linear program, in which the decision variables indicate whether each eligible subfield is selected into the prospective portfolio. In the empirical implementation, the binary linear portfolio problem was solved using PuLP 3.3.0 through the CBC mixed-integer linear programming solver distributed with the package.

For comparison, we define a benchmark portfolio on the same feasibility-screened eligible set $\mathcal{E}_{c,t}$ as a sequential relatedness-based ranking heuristic: candidates are ordered in descending order of contemporaneous relatedness and incorporated one by one until the target condition in Equation (27) is satisfied. This preserves the target-attainment criterion used in the optimization problem while providing a directly comparable heuristic benchmark.

4. Results

This section examines whether the benchmark representation of the research space predicts later structural change, and how these validated signals are used in the prospective portfolio layer. We first assess benchmark predictive performance, the coefficient structure and incremental contribution of the regressors derived from the topology, the robustness of the results to alternative constructions of relative relatedness, and the overall predictive implications of these findings. Then, prospective application uses the retained benchmark specification to compare portfolios for Brazil under alternative target increments and to examine cross-country portfolio shifts in the 2024 setting.

4.1. Benchmark Predictive Performance

The benchmark specification performs strongly in both transition regimes. The in-sample evidence summarized in Table 3 shows that entry reaches $R^2_{\mathrm{cont}} = 0.4796$ and AUC $= 0.8093$, while exit reaches $R^2_{\mathrm{cont}} = 0.7854$ and AUC $= 0.9088$. The global continuous fit is $R^2_{\mathrm{cont}} = 0.6629$. Substantively, this reproduces the familiar asymmetry observed in related-diversification settings: the erosion of established specialization is more predictable than the acquisition of new specialization [6,7].

Table 3. In-sample predictive performance of the benchmark specification (relative relatedness standardized over the opportunity set).

Metric	Entry	Exit	Global
R^2_{cont}	0.4796	0.7854	0.6629
AUC	0.8093	0.9088	–

This signal content is not confined to a small number of favorable years. The leave-one-year-out evidence summarized in Table 4 shows that the benchmark specification remains stable across held-out annual slices. Within this design, the mean LOYO AUC is 0.8093 for entry and 0.9116 for exit, while the corresponding mean R^2_{cont} values are 0.4739 and 0.7955. Even the weakest held-out years remain comfortably above trivial discrimination levels, which suggests that the empirical content of the topology is stable rather than episodic.

Table 4. Summary of leave-one-year-out validation for the benchmark specification (relative relatedness standardized over the opportunity set).

	Entry			Exit		
Metric	**Mean**	**Median**	**Min**	**Mean**	**Median**	**Min**
R^2_{cont}	0.4739	0.4691	0.4355	0.7955	0.7910	0.6918
AUC	0.8093	0.8047	0.7897	0.9116	0.9144	0.8857

4.2. Coefficient Structure and the Incremental Role of Topology-Derived Regressors

The coefficient structure of the benchmark specification provides additional insight into the source of predictive performance. As shown in Table 5, the dominant predictor in both regimes is $r_{c,s,\tau}$, which is consistent with the persistence logic of the economic-complexity literature: past specialization remains the strongest predictor of future specialization. The role of relative relatedness is sharply asymmetric across regimes. In the entry equation, $\widetilde{\omega}^{\mathcal{O}}_{c,s,t}$ is positive on average (0.0602) and highly stable across years (IQR = 0.0106), whereas in the exit equation its average coefficient is essentially zero (-0.0018). By contrast, absolute relatedness density $\omega_{c,s,t}$ is positive in both regimes. This indicates that topology-relative positioning matters primarily for diversification into non-specialized activities, whereas exit is better explained by incumbent persistence and absolute connectedness.

Table 5. Average coefficients for the benchmark specification (relative relatedness standardized over the opportunity set). The last column reports the interquartile range (IQR) of the yearly coefficients.

Model	**Variable**	**Average Coefficient**	**Yearly IQR**
Entry	Intercept	0.1382	0.0134
Entry	$r_{c,s,\tau}$	0.6577	0.0275
Entry	$r_{c,s,t}$	-0.0872	0.0863
Entry	$\omega_{c,s,t}$	0.2048	0.0865
Entry	$\widetilde{\omega}^{\mathcal{O}}_{c,s,t}$	0.0602	0.0106
Entry	R^2_{cont}	0.4848	–
Exit	Intercept	0.0397	0.0328
Exit	$r_{c,s,\tau}$	0.8708	0.0915
Exit	$r_{c,s,t}$	-0.0281	0.1246
Exit	$\omega_{c,s,t}$	0.1553	0.0661
Exit	$\widetilde{\omega}^{\mathcal{O}}_{c,s,t}$	-0.0018	0.0039
Exit	R^2_{cont}	0.8023	–

The same conclusion follows from the nested model comparison. The continuous-fit sequence reported in Table 6 shows that entry fit rises from 0.4575 in $\mathcal{S}_1$ to 0.4799 in $\mathcal{S}_4$, with the most visible gains appearing once topology-derived terms are included. Exit fit changes much less, moving from 0.7816 to 0.7861. Hence, the incremental explanatory value of the topology is concentrated on the emergence of new specialization rather than on the deterioration of already established activities.

Table 6. Nested-model continuous fit.

Specification	**Entry R^2_{cont}**	**Exit R^2_{cont}**
$\mathcal{S}_1$: $r_{c,s,\tau}$	0.4575	0.7816
$\mathcal{S}_2$: $r_{c,s,\tau}, r_{c,s,t}$	0.4579	0.7827
$\mathcal{S}_3$: $r_{c,s,\tau}, r_{c,s,t}, \omega_{c,s,t}$	0.4613	0.7860
$\mathcal{S}_4$: $r_{c,s,\tau}, r_{c,s,t}, \omega_{c,s,t}, \widetilde{\omega}^{\mathcal{O}}_{c,s,t}$	0.4799	0.7861

4.3. Robustness to Alternative Constructions of Relative Relatedness

A remaining concern is whether the benchmark result could be an artifact of a particular way of constructing relative relatedness. To address this, we estimate four variants obtained by crossing two normalization modes (*center* and *z-score*) with two support sets (*all* and *opportunities*). The resulting comparison, reported in Table 7, reveals a high degree of stability. In the exit regime, all four specifications are nearly indistinguishable, both in continuous fit and in LOYO AUC. In the entry regime, the z-score family consistently dominates the centered family, and the *opportunities* support set slightly improves upon the *all* variant.

Table 7. Robustness across alternative constructions of relative relatedness.

Specification	Entry R^2_{cont}	Entry AUC (LOYO)	Exit R^2_{cont}	Exit AUC (LOYO)
Center + all	0.4668	0.7996	0.7853	0.9086
Center + opportunities	0.4673	0.8000	0.7853	0.9087
z-score + all	0.4789	0.8076	0.7854	0.9086
z-score + opportunities	0.4796	0.8083	0.7854	0.9087

This pattern supports using the opportunity-set standardized specification as the benchmark. The choice reflects both empirical performance and conceptual interpretability, not just a large numerical gap. It delivers the strongest entry performance while preserving the most natural interpretation for entry processes, namely, that relative relatedness should be evaluated for the contemporaneous opportunity set. Using a z-score over the opportunity set is methodologically sound and closely aligned with the recommendation in the review literature to employ relative relatedness when comparing across locations or activities [7].

4.4. Synthesis of the Predictive Results

The synthesis below refers to the historical assessment carried out at the country–subfield–year level. The country–year and subfield–year aggregation levels provide the marginal quantities used for volume filtering and RCA normalization; the resulting country–subfield specialization matrix is then used to define portfolios, opportunity sets, ubiquity, proximity and relatedness measures. The entry results refer to country-specific transitions from the opportunity set $\mathcal{O}_{c,t}$ into specialization, as defined by the threshold-crossing event in Equation (13). The results reported above therefore establish three empirical points that motivate the remainder of the analysis:

- At the country–subfield–year level, proximity-derived relatedness measures retain substantial information about subsequent structural change under the benchmark topology-construction protocol.
- That information is not confined to a narrow subset of favorable years, but remains broadly stable under the leave-one-year-out stability check.
- The robustness comparison indicates that the main empirical pattern is not driven by a single arbitrary normalization of relative relatedness. Within that broader pattern, the incremental contribution of $\widetilde{\omega}^{\mathcal{O}}_{c,s,t}$ is concentrated in the entry regime, where the relevant candidates are country-specific subfields not yet specialized at the base year.

We use opportunity-set-standardized relative relatedness as the benchmark specification. This choice is motivated jointly by empirical performance in the entry regime and by conceptual consistency with the opportunity-set interpretation developed above. The next two subsections use this benchmark specification in the prospective portfolio layer, first through an illustrative recent country case and then through a comparative cross-country exercise for the 2024 setting.

4.5. Illustrative Portfolio Comparison for Brazil Under Alternative Target Increments

Figure 3 illustrates the portfolio layer using Brazil in base year 2024. This provides a country–year prospective design case in which each plotted point represents a candidate country–subfield addition evaluated under the same base-year information set. The figure compares candidate portfolios under three target increases in raw portfolio complexity: $\Delta\kappa_{\text{raw}} = 0.10$, 0.15 and 0.20. Each panel shows the eligible candidates after feasibility screening. The horizontal axis reports the auxiliary volume proxy $W^{cp}_{c,s,t}$, and the vertical axis reports predicted future sophistication $\widehat{\text{PCI}}_{s,t+\Delta t}$. Grey points denote all eligible candidates, orange squares denote the benchmark portfolio based on relatedness, and blue circles denote the portfolio selected by optimization. Since $W^{cp}_{c,s,t}$ is used only as a descriptive plotting coordinate, positions toward the upper-left region of the plane correspond to visually more favorable combinations of lower auxiliary stepping-stone volume and higher predicted future sophistication, while the optimization itself uses $\widehat{Y}^{\star}_{c,s,t}$ as the cost term. Because the intermediate matrix at $t + \tau$ is not yet observed for a 2024 base year, this illustration uses the prospective stepping-stone convention defined in Equation (17). In other words, the stepping-stone term is initialized at the observed 2024 specialization level rather than being treated as zero.

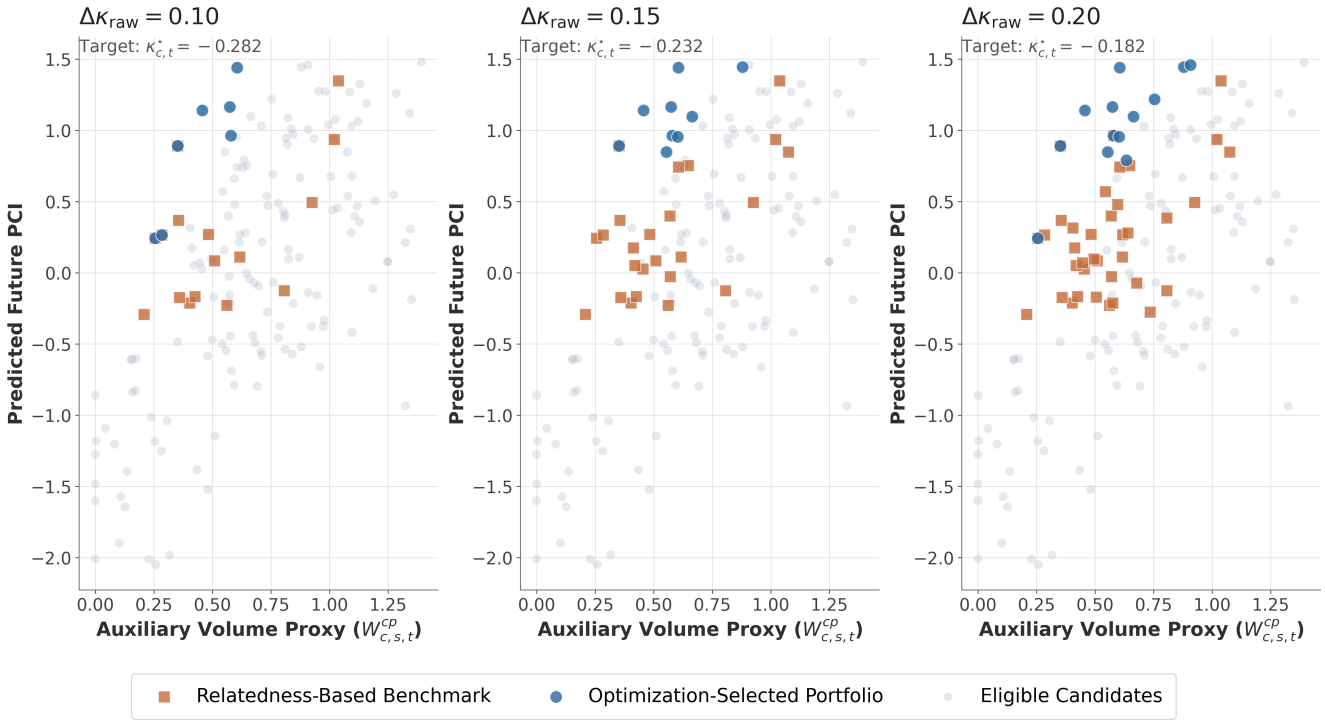


Figure 3. Portfolio based on relatedness and portfolio selected by optimization for Brazil in base year 2024 under alternative target increments in raw portfolio complexity, for $\Delta\kappa_{\text{raw}} = 0.10$, 0.15 and 0.20.

The two procedures diverge across the three target levels. As summarized in Table 8, increasing the target increment changes the size and composition of the selected portfolios:

- For $\Delta\kappa_{\text{raw}} = 0.10$ (left in Figure 3), the benchmark selects 16 activities, whereas the optimization reaches the target with 7.
- For $\Delta\kappa_{\text{raw}} = 0.15$ (middle in Figure 3), the comparison is 24 versus 9.
- For $\Delta\kappa_{\text{raw}} = 0.20$ (right in Figure 3), the benchmark expands to 37 activities, while the optimization requires 13.

Table 8. Target-increment comparison for the Brazil 2024 prospective portfolio-design case.

$\Delta\kappa_{\text{raw}}$	Shared Cand.	Benchmark			Optimized		
		Size	Mean Pred. PCI	Mean W^{cp}	Size	Mean Pred. PCI	Mean W^{cp}
0.10	3	16	0.2374	0.5376	7	0.8718	0.4435
0.15	1	24	0.2817	0.5565	9	1.1039	0.5851
0.20	3	37	0.2553	0.5670	13	1.0495	0.6012

The target-increment comparison in Table 8 indicates that the optimization procedure is not merely trimming the tail of the same relatedness ranking. Across target increments, it constructs materially different combinations of opportunities in order to satisfy the corresponding portfolio-complexity constraint with fewer selected additions. The composition of the selected sets reinforces this interpretation:

- At $\Delta\kappa_{\text{raw}} = 0.10$, only three candidates are shared by both procedures: Ecological Modeling, Reproductive Medicine, and Tourism, Leisure and Hospitality Management. The optimized portfolio instead pivots toward Critical Care and Intensive Care Medicine, Family Practice, Biological Psychiatry, and Endocrine and Autonomic Systems, whereas the benchmark remains anchored in highly related but much lower-sophistication candidates such as Conservation, Linguistics and Language, Applied Microbiology and Biotechnology, and Virology.
- At $\Delta\kappa_{\text{raw}} = 0.15$, the overlap collapses to a single shared activity, Tourism, Leisure and Hospitality Management, while the optimized solution expands into Cell Biology, Software, Otorhinolaryngology, and History and Philosophy of Science.
- By $\Delta\kappa_{\text{raw}} = 0.20$, overlap remains limited to three candidates: Ecological Modeling, Family Practice, and Tourism, Leisure and Hospitality Management. The optimized portfolio then moves further toward candidates with markedly higher predicted future sophistication, including Cellular and Molecular Neuroscience, Clinical Biochemistry, Physical and Theoretical Chemistry, and Biological Psychiatry.

Across the three targets, the optimization therefore departs progressively from pure relatedness ranking and reorients the portfolio toward candidates with substantially stronger predicted future sophistication.

The numerical summaries are consistent with the visual evidence. The mean predicted future PCI among benchmark-selected candidates is 0.2374, 0.2817 and 0.2553 for target increments of 0.10, 0.15 and 0.20, respectively. The corresponding means for the optimized portfolios are 0.8718, 1.1039 and 1.0495. Over the same targets, the average value of $W^{cp}_{c,s,t}$ is 0.5376, 0.5565 and 0.5670 for the benchmark, compared with 0.4435, 0.5851 and 0.6012 for the optimized portfolios. Thus, the optimization clearly dominates the benchmark in predicted future sophistication and, at the two more ambitious targets, does so while accepting only moderate increases in the auxiliary descriptive volume coordinate. Substantively, the optimized portfolios achieve larger predicted sophistication gains than the relatedness heuristic.

The exercise is prospective because it uses the most recent observed portfolio and predicted quantities rather than realized future outcomes. It remains connected to the historical analysis: the optimization cost proxy $\widehat{Y}^{\star}_{c,s,t}$, the plotted auxiliary volume proxy $W^{cp}_{c,s,t}$ and the predicted future sophistication terms $\widehat{\text{PCI}}_{s,t+\Delta t}$ are all generated from the same research-space construction and stepping-stone specification examined in the previous subsections. Thus, the Brazil case extends the historical assessment layer into a targeted portfolio design setting for scholarly diversification.

To complement the metric-based comparison across the three target increments reported in Figure 3 and Table 8, Appendix B provides an illustrative neighborhood-transition view of the Brazil case for the intermediate target increment, $\Delta\kappa_{\text{raw}} = 0.15$. The case uses base year $t = 2024$ and prospective target horizon $t + \Delta t = 2034$. The appendix projects the optimization-selected and relatedness-based portfolios on the retained research-space layout, distinguishing incumbent Brazilian specializations, candidates shared by both procedures, and candidates unique to each procedure. The visual case is intended as an interpretive complement to the analytical comparison reported above; target sensitivity is assessed in the main text through the three values of $\Delta\kappa_{\text{raw}}$.

4.6. Cross-Country Portfolio Shifts in the 2024 Prospective Comparison

To move beyond the single-country illustration, Figures 4 and 5 compare portfolios based on relatedness with portfolios selected by optimization for the 164 countries retained under the baseline 2024 filtering protocol described in Section 3.2.3. The comparison uses the same benchmark prospective setting and rule for meeting the target introduced in Section 3.4. In Figure 4, each country contributes a pair of portfolio outcomes: the portfolio based on relatedness and the corresponding portfolio selected by optimization. The arrows indicate the direction and magnitude of the portfolio shift when the same target is pursued under explicit optimization rather than through a relatedness-ranking heuristic. In turn, Figure 5 aggregates the same comparison by terciles of current country ECI, comprising 55 countries in the high-ECI tercile, 54 in the mid-ECI tercile, and 55 in the low-ECI tercile.

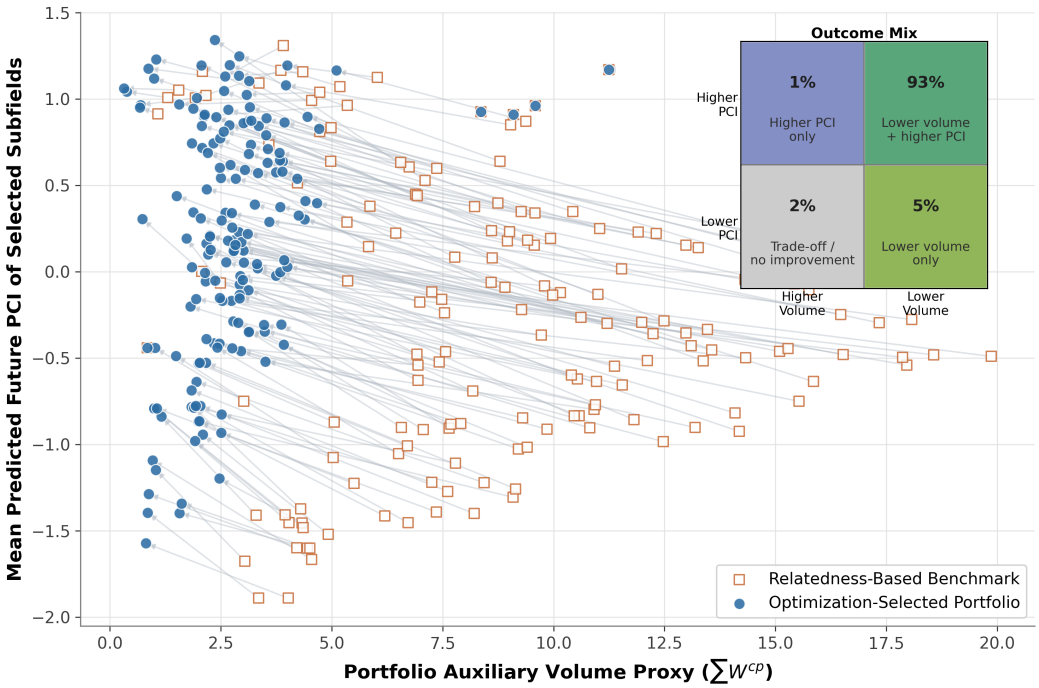


Figure 4. Distribution of portfolio shifts by current ECI tercile in the 2024 prospective comparison.

The aggregate pattern is clear. On average, the relatedness-based portfolios occupy positions with substantially higher auxiliary volume and lower mean predicted future sophistication than their optimization-based counterparts. The mean portfolio auxiliary-volume proxy declines from 19.95 under the relatedness-based benchmark to 7.91 under optimization, while the mean predicted future PCI of selected activities increases from −0.182 to 0.329. Expressed as pairwise portfolio shifts, the mean reduction in the auxiliary volume proxy is 12.03 units (median 11.59), and the mean increase in predicted future PCI is 0.511 (median 0.539). Most importantly, 152 of the 164 countries (92.7%) fall in the

jointly preferred shift category, namely lower auxiliary volume together with higher mean predicted future PCI.

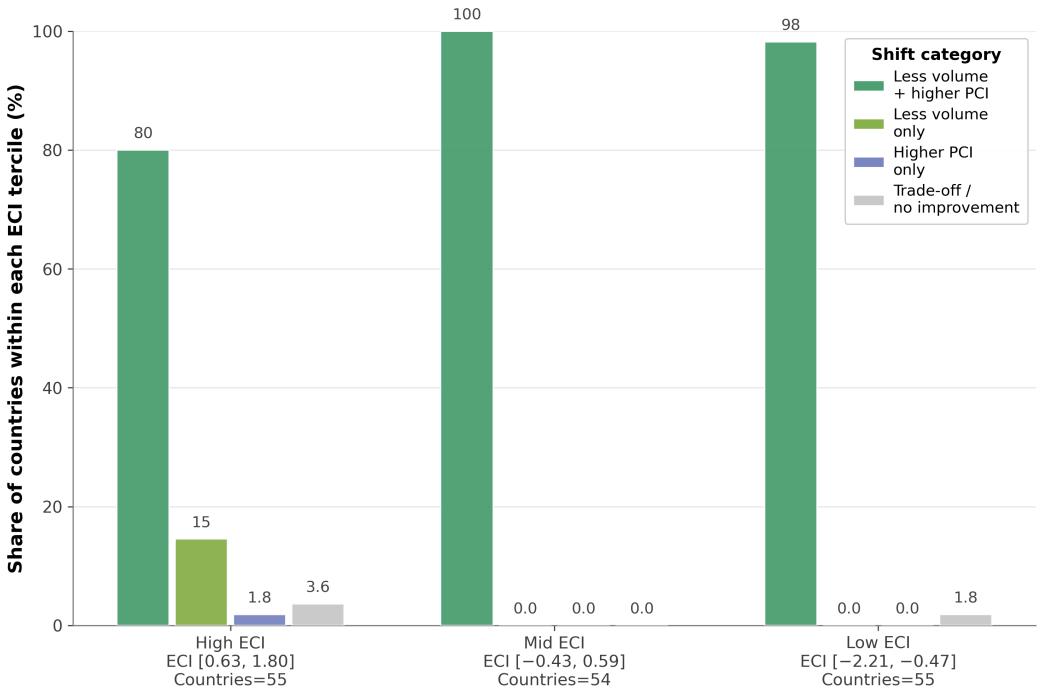


Figure 5. Cross-country distribution of portfolio-shift outcomes by current ECI tercile in the 2024 prospective comparison.

The outcome decomposition in Figure 4 and Table 9 shows that departures from this dominant pattern are limited. Three countries improve only in auxiliary volume, one country improves only in predicted PCI, and eight show no improvement on either dimension. Substantively, this means that the optimization layer does not merely re-rank candidates within a broadly similar opportunity set. In most cases, it produces portfolios that are simultaneously more selective in auxiliary volume terms and more oriented toward higher predicted future sophistication.

Table 9. Summary of portfolio-shift outcomes when moving from the relatedness-ranking benchmark to the optimized portfolio in the 2024 cross-country prospective comparison.

Optimization Outcome Relative to Relatedness Benchmark	Countries	Share (%)
Lower auxiliary volume and higher predicted PCI	152	92.7
Lower auxiliary volume only	3	1.8
Higher predicted PCI only	1	0.6
No improvement on either dimension	8	4.9
Total	164	100.0

Figure 5 shows that the residual heterogeneity is concentrated almost entirely in the high-ECI tercile. Among high-ECI countries (ECI in [0.63, 1.80], $n = 55$), 78.2% still fall into the preferred category of lower auxiliary volume and higher predicted future PCI, but 14.5% fall into the trade-off or no-improvement category, and 3.6% fall into each of the one-dimensional shift categories. Even in this tercile, however, the average shift remains favorable: the mean reduction in auxiliary volume is 8.44, the mean gain in predicted future PCI is 0.364, and the optimization-based portfolios use, on average, 14.6 fewer selected additions than the relatedness-based benchmark.

By contrast, the mid- and low-ECI terciles display a fully uniform directional pattern in this prospective comparison. For mid-ECI countries (ECI in $[-0.43, 0.59]$, $n = 54$), 100% of cases fall into the lower-volume-and-higher-PCI category. The corresponding mean reduction in auxiliary volume is 17.28, the mean increase in predicted future PCI is 0.632, and the optimization-based portfolios use, on average, 23.2 fewer selected additions than the relatedness-based benchmark. The same complete dominance pattern appears in the low-ECI tercile (ECI in $[-2.21, -0.47]$, $n = 55$), where 100% of countries again move toward lower auxiliary volume and higher predicted future PCI, with a mean reduction in auxiliary volume of 10.48, a mean increase in predicted future PCI of 0.539, and an average reduction of 12.3 selected additions.

These cross-country results generalize the country-level illustration by showing that the divergence between heuristic ranking and explicit optimization is not an isolated feature of a single case. Under the benchmark prospective configuration, optimization-based portfolios are typically smaller in auxiliary volume terms, more selective in the number of added activities, and oriented toward substantially higher predicted future sophistication. The fact that the non-dominant cases are concentrated in the high-ECI tercile is also substantively informative: as countries move toward the upper end of the complexity distribution, the opportunity frontier appears tighter and the portfolio problem becomes more heterogeneous, making trade-offs more likely. Even so, the prevailing pattern across the retained 2024 country sample is that optimization-based portfolio design produces lower-volume and higher-sophistication portfolios than relatedness-based sequential selection under the same model-conditional prospective setting.

5. Discussion

Under the retained research-space construction, topology-derived signals contain stable information about subsequent specialization dynamics and remain informative when used in a prospective portfolio-design exercise with an explicit target criterion. The following subsections examine what these findings imply for the historical assessment of the research space, for the differentiated role of topology-derived variables across transition regimes, and for the scope of the prospective portfolio comparisons.

5.1. Historical Assessment of the Benchmark Representation of the Research Space

The central question is whether repeated applications of a common construction protocol yield topology-derived signals that remain informative about later specialization dynamics. Because the underlying activity matrices are computed over rolling windows, the analysis does not assume that the empirical system is literally fixed across time. What remains fixed is the protocol used to derive the research-space representation. The results indicate that repeated applications of that protocol do generate signals with stable historical content, as reflected in both the benchmark fit measures and the year-holdout stability checks reported in Section 4.

This result also clarifies what is established by the historical assessment. In the present setting, a consistently constructed representation of the research space yields topology-derived signals with stable historical content across repeated observations. The robustness exercises strengthen this result by showing that the same empirical pattern persists across reasonable alternative operational choices, rather than being tied to a single construction variant.

A related implication is that the value of the research-space representation in this study lies less in the visual appearance of the network than in the behavior of the quantities derived from it. Prior science-mapping and research-space studies have shown that scientific activity can be represented as structured fields or opportunity spaces [10,11,25,27]. Figure 2

remains useful as a descriptive summary of the retained system, but the main analytical relevance of the representation comes from the historical performance of variables such as $\omega_{c,s,t}$ and $\widetilde{\omega}^{\mathcal{O}}_{c,s,t}$. In this respect, our results extend the mapping role of these representations by evaluating whether topology-derived signals carry usable empirical information under a common historical assessment protocol.

5.2. Entry, Exit and the Role of Relative Relatedness Standardized over the Opportunity Set

A second implication concerns the asymmetric role of topology-derived signals across the two transition regimes. The results show that persistence remains the dominant component in both entry and exit equations, which is consistent with the broader literature on related diversification and structural change [4,6,7]. However, the incremental contribution of the topology-derived terms is not symmetric. Opportunity-set standardized relative relatedness contributes mainly in the entry regime, whereas exit dynamics are more strongly associated with lagged specialization and, to a lesser extent, absolute relatedness density.

Within the logic of the model, this pattern is substantively coherent. Entry is evaluated over the contemporaneous opportunity set, so a relative signal is especially useful because it differentiates among currently absent activities that vary in their fit with the existing portfolio. By contrast, exit concerns activities that are already specialized, for which persistence and incumbent position naturally play a larger role. The empirical results are therefore consistent with the interpretation that relative portfolio position is most informative where the analytical problem is one of prospective expansion into non-specialized activities.

This point further specifies the role of the retained specification in the analysis. Its relevance lies in its correspondence with the structure of the entry problem examined in this paper. This interpretation is consistent with relatedness research showing that the empirical meaning of relatedness depends on the support set and normalization used for comparison [7,17]. In the empirical comparison, opportunity-set standardized relative relatedness delivers the strongest entry-side performance and remains consistent with cross-country comparison over contemporaneous opportunity sets.

5.3. From Historical Assessment to Prospective Portfolio Design

A third implication concerns the relation between the historical assessment layer and the prospective portfolio layer. A frequent difficulty in this line of work is that mapping, predictive assessment and portfolio selection can appear as loosely connected exercises. Our framework reduces that separation by using the historically retained specification to parameterize the portfolio problem. The forward-looking exercise is therefore expressed through quantities whose empirical content has already been examined within the same country–subfield setting.

This point is especially relevant in light of work on relatedness-based prioritization, smart specialization and optimization. In policy-oriented settings, relatedness and complexity are often used to rank promising activities or identify strategic opportunities [32–34]. Recent optimization-oriented work further argues that relatedness-based heuristics and optimization-based selection need not lead to the same diversification choices [15,16]. Our results are consistent with that broader argument. In the 2024 prospective exercises, the optimization-based portfolios differ materially from the relatedness-ranking benchmark under the same target-attainment logic. The Brazil illustration makes this difference visible at the level of candidate composition, while the cross-country comparison shows that the same pattern is not confined to a single case.

The prospective results should be interpreted within the scope of the empirical design. They show how signals validated in the historical analysis can be translated into a portfolio

problem with an explicit target, and how the resulting portfolio can differ from sequential relatedness ranking under the model-implied quantities used in this paper.

5.4. Methodological and Substantive Implications

Methodologically, the article suggests that research-space analysis in scholarly domains becomes more informative when descriptive representation, historical signal assessment and prospective portfolio design are treated as connected stages of a single empirical workflow. Science-mapping and research-space applications have shown how space-based representations can organize fields, visualize scientific structure and support opportunity assessment [10,11,25,27]. Building on this literature, our framework makes explicit how the representation is constructed, how its derived signals are assessed, and how the retained benchmark is used in a forward-looking exercise.

Substantively, the findings indicate that a country's position within the surrounding subfield neighborhood is informative about where future specialization is more likely to emerge. In this setting, the structure of the incumbent portfolio leaves a measurable imprint on the configuration of nearby opportunities. This relationship provides an empirical basis to discuss prospective research expansion at the country level.

For science-policy use, in line with work that uses relatedness, complexity and knowledge-space evidence to inform prioritization [32–34], our framework provides a structured way to identify model-conditioned opportunity patterns under the assumptions and feasibility filters introduced in this paper. The portfolio results thus contribute an empirical basis for examining possible directions of expansion at the country–subfield level alongside institutional, disciplinary and strategic considerations. A stylized country–subfield illustration can make this decision-support interpretation more concrete while preserving the distinction between the empirical framework and any specific policy domain.

Stylized Country–Subfield Policy Illustration

Consider a hypothetical country c whose current specialized portfolio already includes some OpenAlex subfields related to electronic engineering, materials, communication networks, signal processing, instrumentation, environmental engineering, and modeling. Suppose that this country is evaluating whether to expand toward energy-related candidate subfields such as Renewable Energy, Sustainability and the Environment or Energy Engineering and Power Technology. In the framework, these candidates are not treated as broad policy labels, but as OpenAlex subfields whose priority depends on their measured relation to the country's observed portfolio.

Table 10 summarizes how this stylized situation can be interpreted within the country–subfield logic of the framework. The relevant adjacency is not assumed from the energy label itself. It would be estimated from the same country–subfield co-specialization structure used in the empirical analysis. A relatedness heuristic would favor candidate subfields that are close to the country's current research portfolio. The optimization framework, by contrast, would select candidates only if they help satisfy the target portfolio-complexity constraint at comparatively low predicted incorporation effort. The table therefore illustrates how stakeholders could use the framework to compare candidate directions without treating the model output as a deterministic policy prescription.

Table 10. Stylized country–subfield policy illustration for an energy-related setting.

Element	Stylized Interpretation
Decision unit	Hypothetical country *c*.
Incumbent portfolio	Existing specialization in some adjacent OpenAlex subfields, such as Electrical and Electronic Engineering, Electronic, Optical and Magnetic Materials, Computer Networks and Communications, Signal Processing, Instrumentation, Environmental Engineering, and Modeling and Simulation.
Candidate subfields	Energy-related OpenAlex subfields not yet specialized by country *c*, such as Renewable Energy, Sustainability and the Environment, or Energy Engineering and Power Technology.
Adjacency estimate	Estimated from the observed country–subfield portfolio and the co-specialization structure of the research space, rather than assumed from the policy-domain label.
Policy question	Should country *c* prioritize one or more energy-related candidate subfields in its next research strategy?
Relatedness heuristic	Prioritize candidate subfields if they are close to the country's current research portfolio.
Optimization framework	Select candidate subfields only if they contribute to the target portfolio-complexity gain at comparatively low predicted incorporation effort.
Stakeholders	Science ministries, funding agencies, university consortia, energy-transition agencies and research-performing organizations.
Policy instruments	Targeted calls, doctoral networks, shared laboratories, interdisciplinary centers and industry-linked demonstrators.

The energy-related setting is useful here because it resembles the kind of cross-subfield opportunity that the framework is intended to compare. Applied research in this area may combine renewable-energy engineering, thermal modeling, materials science, sensing technologies and low-power electronic systems. Studies on thermal solar collectors using nanofluids, autonomous wireless-sensor nodes powered by thermal energy harvesting and self-powered IoT sensor nodes based on hybrid light–thermal harvesting illustrate this type of technical combination [36–38]. The framework does not prescribe this domain as a priority; rather, it provides a way to evaluate whether the corresponding candidate subfields are sufficiently related to the observed portfolio and whether they contribute to the target-conditioned portfolio-design objective.

6. Conclusions

This study examined whether a research-space representation can move beyond descriptive mapping and support historically assessed, prospective portfolio design for scholarly diversification. The analysis was developed at the country–subfield level, using a common construction protocol to derive topology-based signals, evaluate their historical content and apply the retained specification to a target-based portfolio problem.

The analysis supports three findings:

- Research-space signals show stable predictive value over time. Our results show that the retained research-space construction yields topology-derived signals with stable predictive content across the analytical period. In the present setting, the empirical relevance of the research space lies in the historical behavior of variables derived from it, particularly relatedness density and opportunity-set standardized relative relatedness.

- Entry and exit regimes are associated with different roles for topology-derived variables. Existing specialization levels remain the strongest predictors of subsequent specialization in both regimes, but the incremental contribution of opportunity-set standardized relative relatedness is concentrated in the entry regime. This pattern is consistent with the role of relative portfolio position when the analytical problem concerns expansion into currently non-specialized activities. The retained specification is therefore supported by both comparative performance and its correspondence with the structure of the entry problem.
- Signals retained from the historical analysis can inform target-oriented prospective portfolio design. In the prospective exercises reported here, optimization-based portfolios do not coincide mechanically with relatedness-ranking heuristics and can yield materially different combinations of candidate subfields under the same target-attainment logic. The empirical result is a model-conditioned comparison of alternative portfolio-selection rules generated from the retained historical framework.

Overall, the article provides an empirically grounded framework for scholarly diversification in which research-space signals are historically assessed before being used in prospective portfolio design. The framework is best interpreted as a decision-support structure: it organizes evidence about feasible research expansion under explicit assumptions, but it does not prescribe priorities independently of national, disciplinary or policy judgment.

6.1. Limitations

The following limitations delimit the interpretation of the findings:

- First, the empirical analysis depends on OpenAlex-based publication data. OpenAlex provides broad and reproducible coverage [12], but the resulting country–subfield counts still depend on the underlying indexing, affiliation assignment and field-classification processes. In that sense, the representation of the research space remains conditional on the source from which the scholarly activity matrix is derived.
- Second, the retained representation depends on the construction protocol. Window length, filtering thresholds, minimum ubiquity and diversity requirements, and the RCA-based specialization rule all shape the analytical system examined in this paper. The volume filters are consistent with the matrix logic of economic complexity, where RCA, diversity, ubiquity, proximity and relatedness are computed from country–activity specialization structures [4,5,7]. In the present bibliometric setting, they also reduce the instability associated with very small publication sets [39]. This choice, however, shifts the primary analytical set toward more established research ecosystems. For smaller or emerging countries outside the retained sample, topology-derived signals may be more sensitive to sparse activity, discontinuous publication patterns, or concentration in a narrow subfield portfolio. The robustness exercises support the stability of the main pattern within the retained system, while leaving open whether alternative filtering schemes would yield comparable signals for lower-volume research ecosystems.
- Third, the historical assessment layer is reduced-form. Its coefficients quantify the association between topology-derived signals and subsequent specialization dynamics within the empirical design of the study. This makes the framework suitable for predictive assessment and comparative portfolio analysis at the country–subfield level, while questions of causal attribution call for a different class of empirical design.
- Fourth, the prospective portfolio layer is defined by predicted quantities. The estimated entry-effort proxy $\widehat{Y}^{\star}_{c,s,t}$ and the predicted future sophistication terms $\widehat{\mathrm{PCI}}_{s,t+\Delta t}$ are generated from the retained representation of the research space and transition specification. For recent base years, such as 2024, the portfolio exercise also incor-

porates the carry-forward convention introduced in Equation (17). The resulting comparison therefore concerns prospective portfolio differences under the quantities and conventions specified in the model.

- Finally, the analysis is conducted at the country–subfield level. This level matches the objective of the article and captures broad national diversification structure, while leaving institutional heterogeneity within countries unresolved. The framework thus represents scholarly diversification at the country level, whereas the organizational processes through which research capabilities are accumulated and reconfigured remain a distinct layer of analysis.

6.2. Future Work

A first direction to advance this research would be to apply the same design at other levels of aggregation, especially institutions, regions or mission-oriented research programs. This would help clarify whether the combination of historical signal assessment and target-oriented portfolio design remains informative when the decision unit is more organizationally specific than the country.

It would also be interesting to develop a dynamic research-space framework for studying how academic fields evolve over time. The present study uses repeated country–subfield representations to evaluate specialization dynamics, but it still relies on a fixed subfield taxonomy as the analytical unit. Future work could examine how academic fields change beyond rigid classification structures, by tracing shifts in field positions, proximity relations, community composition, and emergence or decline patterns across rolling windows. This would extend the present portfolio framework toward a longitudinal analysis of academic-field evolution.

The framework could also be extended to smaller and emerging research ecosystems. The present study uses volume filters to obtain stable country–subfield signals, so lower-volume countries are not the primary analytical setting. Future work could test alternative minimum-support rules, longer aggregation windows and bootstrap-based stability diagnostics for proximity and relatedness density. This would help assess whether topology-derived signals remain informative when country-level publication activity is sparse, intermittent or concentrated in a narrow subfield portfolio.

Another possible extension would be to enrich the prospective portfolio layer with additional feasibility dimensions. In this paper, the portfolio problem is parameterized by model-based entry effort and predicted future sophistication. Future work could incorporate further constraints related to funding, staffing, infrastructure, collaboration requirements or strategic policy priorities, thereby bringing the portfolio problem closer to concrete research-management settings.

Finally, we intend to study alternative measurement and representation choices. We have relied on publication-based specialization, RCA-derived presence and an OpenAlex-based subfield taxonomy. Future work could examine the robustness of the main findings under alternative activity measures, alternative field classifications, topic-embedding-based spaces, or hybrid representations that combine co-specialization with other forms of scientific proximity. Beyond robustness assessment alone, these extensions could motivate a many-valued logic for the prospective decision layer. Instead of ranking candidate subfields solely by a single scalar predictive signal, such an approach could classify them as strongly supported, weakly supported, conflicting-signal, indeterminate or unsupported opportunities, thereby offering a more principled way to handle agreement, uncertainty and conflict across alternative representations [40].

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Abbreviations

AUC	Area under the receiver operating characteristic curve
ECI	Economic Complexity Index
IQR	Interquartile Range
LOYO	Leave-One-Year-Out
OEC	Observatory of Economic Complexity
PCI	Product Complexity Index
PR-AUC	Area under the precision–recall curve
RCA	Revealed Comparative Advantage
VIF	Variance Inflation Factor
Mathematical symbols	
Data and matrices	
$X_{c,s}^{(t,h)}$	Aggregated document count for country c in subfield s over a window of size h ending at year t.
$M_{c,s}^{(t,h)}$	Binary specialization indicator: $M_{c,s}^{(t,h)} = 1$ if $\mathrm{RCA}_{c,s}^{(t,h)} \geq 1$, and 0 otherwise.
$\mathrm{RCA}_{c,s}^{(t,h)}$	Revealed comparative advantage of country c in subfield s at window (t, h).
Δt	Forecasting horizon used to evaluate future specialization dynamics.
τ	Intermediate stepping-stone offset used to bridge the transition between the base year and the forecasting horizon.
Complexity and proximity	
$k_{c,0}^{(t,h)}$	Country diversity: number of subfields in which country c is specialized.
$k_{s,0}^{(t,h)}$	Subfield ubiquity: number of countries specialized in subfield s.
$\phi_{s,s'}^{(t,h)}$	Proximity between subfields s and s', defined as the minimum of the two conditional probabilities of co-specialization.
$\omega_{c,s,t}$	Relatedness density of subfield s for country c at year t.
$\mathcal{O}_{c,t}$	Opportunity set of country c at year t, consisting of subfields in which the country is not yet specialized.
$\widetilde{\omega}_{c,s,t}^{\mathcal{O}}$	Opportunity-set standardized relative relatedness, computed as the z-score of $\omega_{c,s,t}$ over $\mathcal{O}_{c,t}$.
Stepping-stone predictive model	
$r_{c,s,t}$	Log-transformed RCA, $r_{c,s,t} = \log(1 + \mathrm{RCA}_{c,s,t})$.
$E_{c,s,t}^{(\Delta t)}$	Entry indicator: threshold-crossing event of RCA from below over horizon Δt.
$X_{c,s,t}^{(\Delta t)}$	Exit indicator: threshold-crossing event of RCA from above over horizon Δt.
$\widehat{r}_{c,s,t+\Delta t}$	Predicted transformed RCA at horizon $t + \Delta t$.
$\widehat{\mathrm{PCI}}_{s,t+\Delta t}$	Predicted subfield complexity index, adapted from the PCI logic, at horizon $t + \Delta t$.

Portfolio-optimization problem	
$\mathcal{S}_{c,t}$	Incumbent specialized set of country c at year t.
$\mathcal{E}_{c,t}$	Final eligible set of candidate subfields for country c at year t, obtained after the feasibility screen.
$\kappa_{c,t}^{\mathrm{raw}}$	Raw portfolio complexity for country c at year t.
$\kappa_{c,t}^{\star}$	Target portfolio complexity for country c at year t.
$z_{c,s}$	Binary decision variable: $z_{c,s} = 1$ if subfield s is selected in the optimized portfolio of country c.
$\widehat{Y}_{c,s,t}^{\star}$	Estimated entry-effort proxy for candidate subfield s in country c at year t.
Glossary of recurrent technical terms	
Research space	Network of subfields connected by patterns of co-specialization across countries.
Relative relatedness	Standardized version of relatedness density computed over the opportunity set.
Proximity	Symmetric similarity between two subfields, defined as the minimum of the two conditional probabilities of co-specialization.
Stepping-stone observation	Intermediate time point used to bridge the transition over horizon Δt in the predictive model.
Opportunity set	Set of subfields in which a country is not yet specialized at year t.
Portfolio complexity	Mean PCI of the subfields included in a country's specialized or optimized portfolio, depending on the context.
Entry regime	Event in which a country's RCA crosses the specialization threshold from below over horizon Δt.
Exit regime	Event in which a country's RCA crosses the specialization threshold from above over horizon Δt.
Subfield	OpenAlex subject classification used as the analytical unit of scholarly activity.

Appendix A. Topology Robustness and Alternative Filter Choices

This appendix reports the supplementary evidence used to assess the robustness of the retained research-space topology. The material follows the same analytical progression as the construction protocol discussed in the main text. It begins with the composition of the retained sample, then examines the sensitivity of the proximity structure to alternative operational choices, and concludes with compact comparisons across threshold families, aggregation windows and filtering modes.

Table A1 reports the restriction from the raw universe to the retained filtered sample under the baseline configuration. The table extends the same filtering protocol across the full period used in the manuscript, from the benchmark years to the most recent prospective base year. Its purpose is to show how the volume-based filtering step reduces the number of countries, subfields and active country–subfield cells while preserving most of the observed research mass.

Table A2 extends the robustness analysis from sample composition to network structure. For selected years, it compares the retained benchmark against alternative threshold families, diversity and ubiquity cutoffs, and proximity variants.

Table A3 provides a compact threshold-scenario summary for year 2004. Unlike Table A2, which focuses on structural similarity metrics, this table concentrates on the size of the retained sample and the amount of research mass preserved under low, baseline and high threshold variants centered on three threshold families.

Table A4 isolates the effect of the aggregation window by comparing proximities obtained under $h = 5$ and $h = 10$ for selected years. This comparison supports the use of the longer window for visualization by showing that the broad rank structure of the retained topology remains close to the inference-window construction.

Table A1. Baseline sample restriction for benchmark and prospective years ($h = 5$; $thr_C = 1250$, $thr_S = 250$, $thr_{CS} = 50$), reporting the number of countries, subfields and cells before and after filtering, together with the retained research mass.

Year	Countries Before	Countries After	Subfields Before	Subfields After	Cells Before	Cells After	Retained Mass
2004	224	101	251	248	22,860	8730	0.9782
2005	228	102	252	249	23,537	9109	0.9797
2006	231	105	252	249	24,299	9535	0.9811
2007	230	108	252	249	24,958	9975	0.9821
2008	231	114	252	250	25,459	10,434	0.9834
2009	234	121	252	250	26,251	10,967	0.9846
2010	236	124	252	250	27,044	11,500	0.9856
2011	237	126	252	250	27,766	11,960	0.9861
2012	237	131	252	251	28,718	12,518	0.9867
2013	239	137	252	251	29,604	13,108	0.9871
2014	237	142	252	251	30,316	13,734	0.9876
2015	239	145	252	251	30,960	14,293	0.9879
2016	238	148	252	251	31,500	14,879	0.9883
2017	238	149	252	251	32,210	15,422	0.9885
2018	235	150	252	251	32,846	15,976	0.9890
2019	240	152	252	251	33,497	16,470	0.9893
2020	240	157	252	252	34,347	17,120	0.9899
2021	238	159	252	252	35,021	17,803	0.9905
2022	242	159	252	252	35,190	18,293	0.9909
2023	240	163	252	252	35,379	18,774	0.9913
2024	241	164	252	252	34,927	18,975	0.9914

Table A2. Selected-year proximity sensitivity under threshold and filter variants. Metrics are reported relative to the baseline ($ub = 3$, $dv = 3$, minimum conditional-probability proximity).

Year	(thrC, thrS, thrCS)	ub	dv	prox	Spearman	jacc_topk	jacc_mst	c_{raw}	s_{raw}	c_{use}	s_{use}	Density	Edges_pos	mst_Edges
2004	(1250, 250, 50)	1	3	max	1.0000	0.9798	0.9636	101	248	98	248	0.2062	28,931	247
2004	(1250, 250, 50)	3	1	max	1.0000	0.9964	1.0000	101	248	101	239	0.2071	27,612	238
2004	(1250, 250, 50)	3	3	max	1.0000	1.0000	1.0000	101	248	98	239	0.2132	27,612	238
2004	(1250, 250, 50)	3	3	sqrt	0.9398	0.6764	0.4381	101	248	98	239	0.2132	27,612	238
2004	(1250, 250, 50)	3	5	max	0.9999	0.9857	0.9669	101	248	95	239	0.2194	27,612	238
2004	(1250, 250, 50)	5	3	max	1.0000	0.9550	0.9292	101	248	98	226	0.2233	25,050	225
2004	(2500, 500, 100)	3	3	max	0.8838	0.5504	0.1793	77	243	75	230	0.2303	25,539	229
2004	(625, 125, 25)	3	3	max	0.9153	0.6145	0.2559	123	250	119	244	0.2126	28,784	243
2009	(1250, 250, 50)	1	3	max	1.0000	0.9877	0.9522	121	250	116	250	0.2146	29376	249
2009	(1250, 250, 50)	3	1	max	1.0000	0.9889	0.9835	121	250	121	242	0.2123	28,233	241
2009	(1250, 250, 50)	3	3	max	1.0000	1.0000	1.0000	121	250	116	242	0.2211	28,233	241
2009	(1250, 250, 50)	3	3	sqrt	0.9376	0.6639	0.4094	121	250	116	242	0.2211	28,233	241
2009	(1250, 250, 50)	3	5	max	0.9998	0.9853	0.9593	121	250	109	242	0.2344	28,233	241
2009	(1250, 250, 50)	5	3	max	1.0000	0.9806	0.9668	121	250	116	234	0.2276	26,707	233
2009	(2500, 500, 100)	3	3	max	0.9129	0.5959	0.1875	97	249	93	235	0.2282	26,601	234
2009	(625, 125, 25)	3	3	max	0.9428	0.6714	0.2756	140	251	139	246	0.2182	29,426	245
2014	(1250, 250, 50)	1	3	max	1.0000	0.9936	0.9840	142	251	142	251	0.2107	30,036	250
2014	(1250, 250, 50)	3	1	max	1.0000	1.0000	1.0000	142	251	142	247	0.2139	29,524	246
2014	(1250, 250, 50)	3	3	max	1.0000	1.0000	1.0000	142	251	142	247	0.2139	29,524	246
2014	(1250, 250, 50)	3	3	sqrt	0.9428	0.6639	0.4643	142	251	142	247	0.2139	29,524	246
2014	(1250, 250, 50)	3	5	max	0.9999	0.9893	0.9680	142	251	136	247	0.2227	29,524	246
2014	(1250, 250, 50)	5	3	max	1.0000	0.9944	0.9797	142	251	142	242	0.2178	28,580	241
2014	(2500, 500, 100)	3	3	max	0.9294	0.6171	0.2261	118	249	117	243	0.2169	28,362	242
2014	(625, 125, 25)	3	3	max	0.9396	0.6609	0.3397	159	251	158	248	0.2289	30,000	247

Table A3. Threshold-scenario summary for year 2004 ($h = 5$) under three scenario families centered on baseline cell thresholds $thr_{CS} \in \{40, 50, 60\}$. Within each family, the low, baseline and high rows correspond to proportional threshold variants.

Threshold Family	Scenario	thr_C	thr_S	thr_{CS}	$n_{\text{countries}}$	$n_{\text{subfields}}$	Retained_Mass
thrCS40	low	625	125	20	123	250	0.9908
thrCS40	baseline	1250	250	40	101	249	0.9821
thrCS40	high	2500	500	80	77	247	0.9661
thrCS50	low	625	125	25	123	250	0.9885
thrCS50	baseline	1250	250	50	101	248	0.9782
thrCS50	high	2500	500	100	77	243	0.9593
thrCS60	low	625	125	30	123	249	0.9864
thrCS60	baseline	1250	250	60	100	248	0.9742
thrCS60	high	2500	500	120	77	243	0.9529

Table A4. Window sensitivity of proximities comparing $h = 5$ and $h = 10$ for selected years.

Year	Spearman ϕ ($h = 5$ vs. $h = 10$)	Jaccard_topk ($h = 5$ vs. $h = 10$)
2004	1.0000	1.0000
2009	0.9518	0.7376
2014	0.9624	0.7528

Table A5 compares the filtered sample with the unfiltered universe for selected years. Its purpose is to show more directly how the retained topology differs from the topology that would emerge without the volume-based restriction, thereby clarifying why the filtered construction is treated as the analytical system of record in this paper.

Table A5. Mode sensitivity comparing the unfiltered universe with the filtered sample (baseline thresholds, $h = 5$, $thr_{CS} = 50$) for selected years.

Year	$n_{\text{subfields}}$ Common	Spearman ϕ (Universe vs. Filtered)	Jaccard_topk (Universe vs. Filtered)
2004	239	0.6846	0.3837
2009	242	0.7293	0.4053
2014	247	0.7317	0.4132

Appendix B. Brazil Neighborhood-Transition Case

This appendix provides a visual complement to the Brazil portfolio illustration reported in Section 4.5. The purpose is to show how the optimized and relatedness-based portfolios for base year $t = 2024$, target horizon $t + \Delta t = 2034$, and $\Delta\kappa_{\text{raw}} = 0.15$ are positioned relative to Brazil's incumbent research neighborhood in the retained research space. The underlying research-space construction follows the procedure defined in Section 3.2, while the visual layout is used only for orientation, in the same spirit as Figure 2. Node positions and visible edges are not used by the optimization procedure; the portfolio results in the main text are based on the quantitative metrics, feasibility screen and target constraint defined in Section 3.4.

Figures A1 and A2, together with Table A6, reinforce the distinction observed in the main Brazil comparison for $\Delta\kappa_{\text{raw}} = 0.15$ (in both figures, orange nodes denote subfields in which Brazil is specialized in 2024, red nodes denote candidates selected only by optimization, purple nodes denote candidates shared with the relatedness benchmark, and grey nodes denote other retained subfields. The layout represents the retained research space for visual orientation only). The optimization-selected portfolio is more compact,

with nine candidates, whereas the relatedness benchmark selects twenty-four candidates. In the optimization-selected portfolio, all selected additions have at most two incumbent neighbors in the full proximity graph. By contrast, the relatedness benchmark includes a longer tail of candidates with several incumbent neighbors. This pattern is consistent with the interpretation that the benchmark follows more local extensions of the incumbent neighborhood, whereas the optimization procedure selects a smaller set of additions that satisfies the target-complexity condition under the specified effort criterion.



Figure A1. Brazil neighborhood-transition view for the optimization-selected portfolio, using base year $t = 2024$, target horizon $t + \Delta t = 2034$, and $\Delta\kappa_{\text{raw}} = 0.15$.

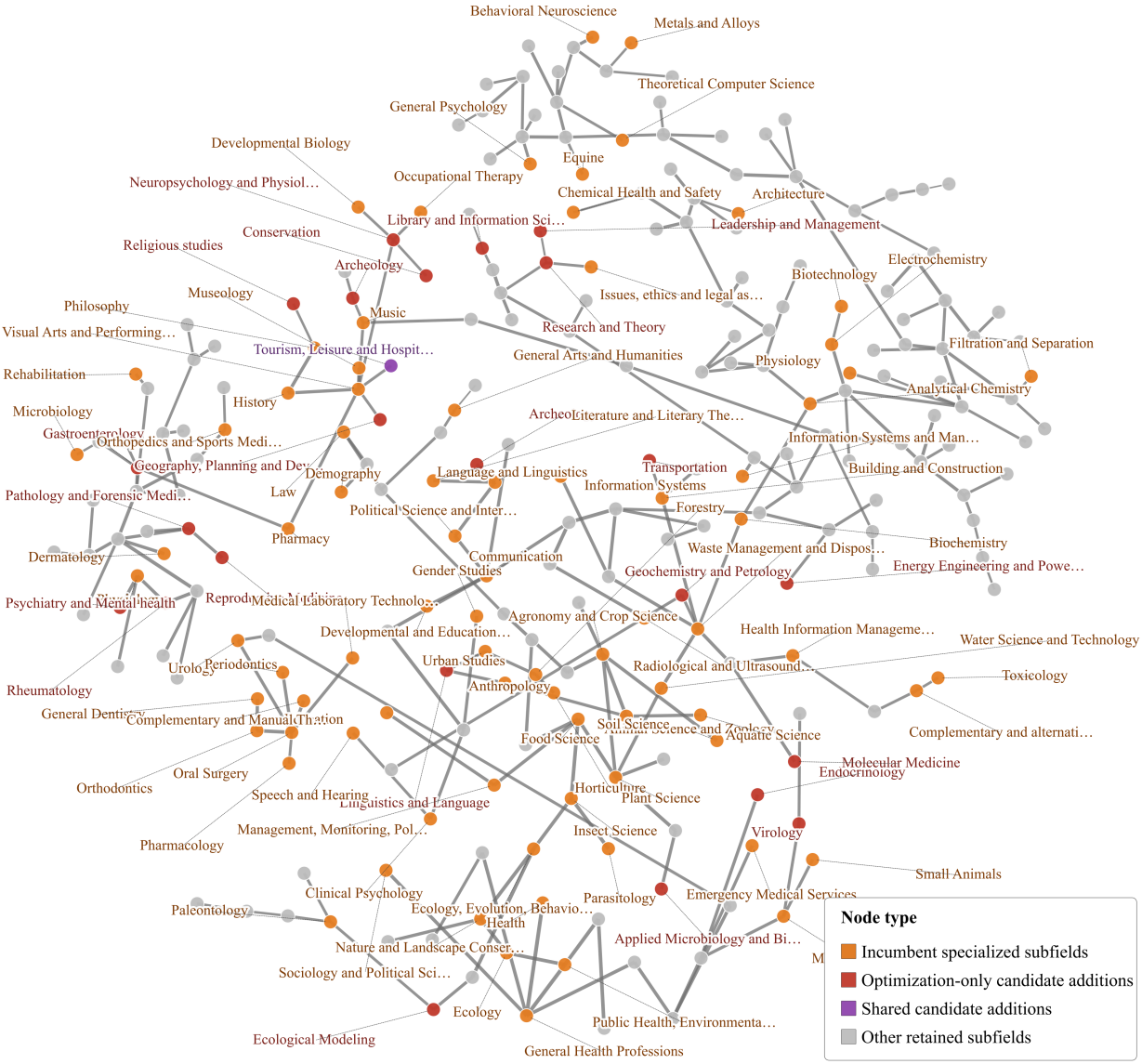


Figure A2. Brazil neighborhood-transition view for the relatedness-based benchmark, using base year $t = 2024$, target horizon $t + \Delta t = 2034$, and $\Delta\kappa_{\text{raw}} = 0.15$.

Table A6. Distribution of incumbent-neighbor counts for Brazil candidates selected under optimization and the relatedness benchmark for $\Delta\kappa_{\text{raw}} = 0.15$. Counts are computed on the full 2024 proximity graph used for the neighborhood analysis.

Incumbent Neighbors	Optimization Count	Relatedness Count	Optimization Share (%)	Relatedness Share (%)
0	3	4	33.3	16.7
1	5	6	55.6	25.0
2	1	1	11.1	4.2
3	0	4	0.0	16.7
4	0	1	0.0	4.2
5	0	1	0.0	4.2
6	0	1	0.0	4.2
8	0	1	0.0	4.2
9	0	1	0.0	4.2
10	0	1	0.0	4.2
12	0	1	0.0	4.2
14	0	1	0.0	4.2
20	0	1	0.0	4.2

Appendix C. Supplementary Predictive Diagnostics

This appendix reports supplementary diagnostics for the retained predictive specification. These materials complement the continuous-fit and AUC evidence discussed in the main text by adding threshold-based classification summaries, robustness comparisons across alternative relative-relatedness constructions, specification diagnostics and placebo benchmarks. The tables are ordered from predictive classification summaries to robustness comparisons and then to model-diagnostic evidence.

Table A7 reports threshold-based classification summaries for the retained specification. Its role is to complement the continuous-fit and AUC results with a direct view of precision–recall trade-offs in the entry and exit regimes.

Table A7. Supplementary threshold-based diagnostics for the benchmark specification (relative relatedness standardized over the opportunity set).

Scope	Regime	Precision	Recall	F_1
In-sample	Entry	0.7707	0.4593	0.5756
In-sample	Exit	0.7894	0.5619	0.6565
LOYO mean	Entry	0.7719	0.4616	0.5766
LOYO mean	Exit	0.7923	0.5662	0.6592

Table A8 compares PR-AUC across the alternative constructions of relative relatedness examined in the robustness analysis. This table is useful because PR-AUC remains informative when class imbalance matters and when ranking quality is of interest beyond the AUC values reported in the main text.

Table A8. PR-AUC across robustness specifications.

Specification	Entry PR-AUC (In-Sample)	Entry PR-AUC (LOYO)	Exit PR-AUC (In-Sample)	Exit PR-AUC (LOYO)
Center + all	0.6177	0.6168	0.7406	0.7401
Center + opportunities	0.6174	0.6165	0.7406	0.7401
Z-score + all	0.6144	0.6138	0.7403	0.7399
Z-score + opportunities	0.6148	0.6141	0.7405	0.7401

Table A9 reports pseudo-Brier scores for the same robustness variants. Its role is to complement the ranking-oriented metrics with a summary of probabilistic error under the alternative relative-relatedness constructions compared in this paper.

Table A9. Pseudo-Brier scores across robustness specifications.

Specification	Entry Brier (In-Sample)	Entry Brier (LOYO)	Exit Brier (In-Sample)	Exit Brier (LOYO)
Center + all	0.1762	0.1762	0.1994	0.1993
Center + opportunities	0.1762	0.1763	0.1994	0.1993
Z-score + all	0.1756	0.1757	0.1994	0.1994
Z-score + opportunities	0.1756	0.1756	0.1994	0.1994

Table A10 reports pairwise correlations and variance inflation factors for the regressors in the retained specification. This table clarifies the degree of overlap among the main explanatory terms and shows that the retained model is not driven by extreme multicollinearity.

Table A10. Pairwise correlations and variance inflation factors for the benchmark specification.

Statistic	Entry	Exit
$\mathrm{corr}(\widetilde{\omega}^{\mathcal{O}}_{c,s,t}, \omega_{c,s,t})$	0.2564	0.0171
$\mathrm{corr}(\widetilde{\omega}^{\mathcal{O}}_{c,s,t}, r_{c,s,t})$	0.2576	0.2016
$\mathrm{corr}(\widetilde{\omega}^{\mathcal{O}}_{c,s,t}, r_{c,s,\tau})$	0.3727	0.2030
$\mathrm{corr}(\omega_{c,s,t}, r_{c,s,t})$	0.7052	−0.5031
$\mathrm{corr}(\omega_{c,s,t}, r_{c,s,\tau})$	0.4119	−0.3373
$\mathrm{corr}(r_{c,s,t}, r_{c,s,\tau})$	0.4062	0.8550
$\mathrm{VIF}(\widetilde{\omega}^{\mathcal{O}}_{c,s,t})$	1.1833	1.0643
$\mathrm{VIF}(\omega_{c,s,t})$	2.0725	1.4232
$\mathrm{VIF}(r_{c,s,t})$	2.0628	4.6625
$\mathrm{VIF}(r_{c,s,\tau})$	1.3582	3.8882

Table A11 reports placebo distributions for the retained benchmark. Its purpose is to show that the predictive performance observed in the historical assessment remains materially above chance levels under randomized benchmarks.

Table A11. Placebo summary for the benchmark specification (relative relatedness standardized over the opportunity set, 200 iterations).

Regime	Metric	Mean	p05	p95
Entry	Placebo AUC	0.4999	0.4979	0.5023
Entry	Placebo PR-AUC	0.1608	0.1599	0.1618
Exit	Placebo AUC	0.4999	0.4961	0.5041
Exit	Placebo PR-AUC	0.2110	0.2086	0.2132

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